

# 第 2 章 資料清理與轉換(tidyverse)

## Table of Contents

|                                                             |    |
|-------------------------------------------------------------|----|
| 大綱:.....                                                    | 4  |
| ❁ 2.1 tidy data 概念與常見資料問題(缺值、重複、型態) .....                   | 4  |
| Tidy Data 簡介 .....                                          | 4  |
| Tidy Data 範例 .....                                          | 4  |
| 不符合 Tidy Data 範例 .....                                      | 5  |
| 資料品質(Data Quality) .....                                    | 5  |
| 1.遺漏值(Missing Values) .....                                 | 6  |
| 填補遺漏值.....                                                  | 11 |
| 方法一：平均數.....                                                | 11 |
| 方法二：中位數.....                                                | 12 |
| 方法三：固定值.....                                                | 12 |
| 2.重複值(Duplicate Records) .....                              | 12 |
| 找出重複資料列.....                                                | 12 |
| 查看重複資料 .....                                                | 12 |
| 刪除重複資料列.....                                                | 12 |
| 刪除重複資料列 {dplyr}.....                                        | 12 |
| 3.資料型態錯誤(Incorrect Data Types) .....                        | 13 |
| 常見資料型態錯誤 .....                                              | 13 |
| 字串轉換為日期.....                                                | 13 |
| 字串轉換為因子 (Factor) .....                                      | 13 |
| 因子轉換字串 .....                                                | 14 |
| ❁ 2.2 dplyr : filter/select/mutate/summarise/group_by ..... | 14 |

|                                                   |    |
|---------------------------------------------------|----|
| 管線運算子(%>%).....                                   | 14 |
| 認識 dplyr 套件.....                                  | 15 |
| flights 2013 年 NYC 機場航班資料, 33 萬筆資料.....           | 15 |
| dplyr 套件重要函數.....                                 | 20 |
| filter 條件式篩選資料.....                               | 20 |
| slice 列的指標篩選資料.....                               | 20 |
| arrange 排序.....                                   | 21 |
| select 選取行.....                                   | 21 |
| select 選取行, 使用冒號(:)表示連續範圍.....                    | 21 |
| select 選取行, 使用負號(-)表示刪除.....                      | 21 |
| select 選取行並且更改欄位名稱.....                           | 21 |
| select 選取行, 使用 starts_with 等參數.....               | 21 |
| rename 更改欄位名稱.....                                | 21 |
| distinct 選取不重覆資料.....                             | 21 |
| mutate 新增欄位.....                                  | 22 |
| summarise - 計算加總.....                             | 24 |
| summarise - 平均起飛延誤.....                           | 24 |
| summarise - 多個統計量.....                            | 24 |
| group_by + summarise.....                         | 25 |
| group_by 多群組 + summarise.....                     | 25 |
| 計算航空公司的平均延誤時間, 由大至小排序.....                        | 25 |
| dplyr in R Cheat Sheet.....                       | 25 |
| 熟悉開放資料集於 dplyr 應用.....                            | 25 |
| ✿ 2.3 tidyr : pivot_longer/pivot_wider、分欄/合欄..... | 26 |
| 長寬資料簡介.....                                       | 26 |

|                                            |    |
|--------------------------------------------|----|
| tidyr + ggplot2 -> 使用 tidyverse 整合套件.....  | 28 |
| 近十年國防預算結構分析 .....                          | 31 |
| 長資料(1 行軍事投資,作業維持,人員維持,其他)轉換為寬資料(4 行) ..... | 33 |
| 方法 1 不使用管線操作 .....                         | 33 |
| 方法 2 使用%>% .....                           | 34 |
| 國防預算結構統計圖.....                             | 34 |
| ✿ 2.4 合併資料：join 家族與鍵值設計 .....              | 35 |
| 合併簡介 .....                                 | 35 |
| 範例 1 left_join - 左表新增右表的 1 個欄位 .....       | 36 |
| 範例 2 left_join - 左表新增右表的多個欄位 .....         | 37 |
| 範例 3 left_join - 右表的欄位找不到.....             | 37 |
| ✿ 2.5 練習：完成一份可分析的乾淨資料集.....                | 38 |
| 參考 Open Data 等資料，完成 Tidy Data 。 .....      | 38 |
| 熟悉以下資料集之資料操作 .....                         | 38 |
| 1. AirPassengers 空中旅客人數資料集 .....           | 38 |
| 2. Titanic 1912 年鐵達尼號資料集.....              | 39 |
| 3. airquality 1973 年紐約空氣品質資料集 .....        | 46 |
| 4. iris 鳶尾花資料集 .....                       | 48 |
| 5. mtcars 汽車資料集.....                       | 50 |
| 6. penguins 南極洲帕爾默站企鵝.....                 | 53 |

```

# file      : hands_on_r_02_tidyverse_notebook.Rmd
# title     : R 入門實作班
# subtitle  : 2. 資料清理與轉換(tidyverse)
# author    : Ming-Chang Lee
# date      : 2026.08.09
# YouTube   : https://www.youtube.com/@aLan9956
# RWEPA     : http://rwepa.blogspot.tw/

```

```
# GitHub : https://github.com/rwepa
# Email : alan9956@gmail.com
```

大綱:

- 2.1 tidy data 概念與常見資料問題(缺值、重複、型態)
- 2.2 dplyr : filter/select/mutate/summarise/group\_by
- 2.3 tidyr : pivot\_longer/pivot\_wider、分欄/合欄
- 2.4 合併資料 : join 家族與鍵值設計
- 2.5 練習 : 完成一份可分析的乾淨資料集

## 🌸 2.1 tidy data 概念與常見資料問題(缺值、重複、型態)

### Tidy Data 簡介

Tidy Data(整潔資料)是由資料科學 Hadley Wickham 提出的資料整理原則，也是 tidyverse 生態系的核心概念。

符合 Tidy Data 的資料，具有三項基本原則：

- 1.每個變數(Variable)使用一個欄(Column)。
- 2.每筆觀測(Observation)使用一行(Row)。
- 3.每個資料值(Value)使用一個儲存格(Cell)。

### Tidy Data 範例

```
library(tibble)

student <- tibble(
  ID = c(1, 2, 3, 4),
  Name = c("ALAN", "BOB", "Cathy", "David"),
  Gender = c("F", "M", "F", "M"),
  Math = c(95, 70, 80, 90),
  English = c(80, 85, 90, 77)
)

print(student)

## # A tibble: 4 × 5
##       ID Name  Gender Math English
##   <dbl> <chr> <chr>  <dbl>   <dbl>
## 1     1  ALAN    F      95      80
## 2     2   BOB    M      70      85
```

|      |           |    |    |
|------|-----------|----|----|
| ## 3 | 3 Cathy F | 80 | 90 |
| ## 4 | 4 David M | 90 | 77 |

此資料廣義而言符合 Tidy Data 原則：

- ID：變數。
- Name：變數。
- Gender：變數。
- Math：變數。
- English：變數。
- 每一列代表一位學生。
- 每個儲存格只有一個值。

## 不符合 Tidy Data 範例

上述範例嚴格來說，不完全符合 Tidy Data 的概念。

- 1.每一欄代表一個變數（Variable），不符合 Tidy 原則，Math、English 其實都是同一個變數「科目」的不同值。
- 2.每一列代表一筆觀測值（Observation），不符合 Tidy 原則，每位學生有兩筆成績，但目前只佔一列。
- 3.每一個儲存格只有一個值（Value），完全符合，每個儲存格只有一個資料值。

上例中的 Math, English 修改為 2 個欄位 (Subject, Score)，修改後為長資料(Long data)，修改後結果如下表所示。

```
# > student_long
# # A tibble: 8 × 5
#   ID Name Gender Subject Score
#   <dbl> <chr> <chr>   <chr>   <dbl>
# 1     1 ALAN  F      Math     95
# 2     1 ALAN  F      English  80
# 3     2 BOB   M      Math     70
# 4     2 BOB   M      English  85
# 5     3 Cathy F      Math     80
# 6     3 Cathy F      English  90
# 7     4 David M      Math     90
# 8     4 David M      English  77
```

## 資料品質(Data Quality)

實務上，資料分析約 60%~80% 的時間都花在資料前處理(Data Cleaning)，常見問題包括：

1.遺漏值(Missing Values) 。 2.重複值(Duplicate Records) 。 3.資料型態錯誤 (Incorrect Data Types) 。

## 1.遺漏值(Missing Values)

- 遺漏值在 R 中顯示為 NA, 不是空白.

```
data(Cars93, package = "MASS")
df <- Cars93
print(head(df))
```

| ##   | Manufacturer | Model   | Type    | Min.Price | Price | Max.Price | MPG.city | MPG.highway |
|------|--------------|---------|---------|-----------|-------|-----------|----------|-------------|
| ## 1 | Acura        | Integra | Small   | 12.9      | 15.9  | 18.8      | 25       | 31          |
| ## 2 | Acura        | Legend  | Midsize | 29.2      | 33.9  | 38.7      | 18       | 25          |
| ## 3 | Audi         | 90      | Compact | 25.9      | 29.1  | 32.3      | 20       | 26          |
| ## 4 | Audi         | 100     | Midsize | 30.8      | 37.7  | 44.6      | 19       | 26          |
| ## 5 | BMW          | 535i    | Midsize | 23.7      | 30.0  | 36.2      | 22       | 30          |
| ## 6 | Buick        | Century | Midsize | 14.2      | 15.7  | 17.3      | 22       | 31          |

| ##   | AirBags            | DriveTrain | Cylinders | EngineSize | Horsepower | RPM  |
|------|--------------------|------------|-----------|------------|------------|------|
| ## 1 | None               | Front      | 4         | 1.8        | 140        | 6300 |
| ## 2 | Driver & Passenger | Front      | 6         | 3.2        | 200        | 5500 |
| ## 3 | Driver only        | Front      | 6         | 2.8        | 172        | 5500 |
| ## 4 | Driver & Passenger | Front      | 6         | 2.8        | 172        | 5500 |
| ## 5 | Driver only        | Rear       | 4         | 3.5        | 208        | 5700 |
| ## 6 | Driver only        | Front      | 4         | 2.2        | 110        | 5200 |

| ##   | Rev.per.mile | Man.trans.avail | Fuel.tank.capacity | Passengers | Length | Wheelbase |
|------|--------------|-----------------|--------------------|------------|--------|-----------|
| ## 1 | 2890         | Yes             | 13.2               | 5          | 177    | 102       |
| ## 2 | 2335         | Yes             | 18.0               | 5          | 195    | 115       |
| ## 3 | 2280         | Yes             | 16.9               | 5          | 180    | 102       |
| ## 4 | 2535         | Yes             | 21.1               | 6          | 193    | 106       |
| ## 5 | 2545         | Yes             | 21.1               | 4          | 186    | 109       |
| ## 6 | 2565         | No              | 16.4               | 6          | 189    | 105       |

| ##   | Width | Turn.circle | Rear.seat.room | Luggage.room | Weight | Origin  | Make          |
|------|-------|-------------|----------------|--------------|--------|---------|---------------|
| ## 1 | 68    | 37          | 26.5           | 11           | 2705   | non-USA | Acura Integra |
| ## 2 | 71    | 38          | 30.0           | 15           | 3560   | non-USA | Acura Integra |

```

a Legend
## 3      67      37      28.0      14      3375 non-USA
  Audi 90
## 4      70      37      31.0      17      3405 non-USA
  Audi 100
## 5      69      39      27.0      13      3640 non-USA
  BMW 535i
## 6      69      41      28.0      16      2880      USA Buick
  Century

# 資料類別: data.frame 資料框物件
class(df)

## [1] "data.frame"

# 資料結構, 包括Factor 因子(資料型態是字元 character), num 數值, int 整數.
str(df)

## 'data.frame':    93 obs. of  27 variables:
## $ Manufacturer      : Factor w/ 32 levels "Acura","Audi",...: 1 1 2
## $ Model             : Factor w/ 93 levels "100","190E","240",...: 49
## $ Type              : Factor w/ 6 levels "Compact","Large",...: 4 3
## $ Min.Price         : num  12.9 29.2 25.9 30.8 23.7 14.2 19.9 22.6
## $ Price             : num  15.9 33.9 29.1 37.7 30 15.7 20.8 23.7 26.
## $ Max.Price         : num  18.8 38.7 32.3 44.6 36.2 17.3 21.7 24.9
## $ MPG.city          : int   25 18 20 19 22 22 19 16 19 16 ...
## $ MPG.highway       : int   31 25 26 26 30 31 28 25 27 25 ...
## $ AirBags           : Factor w/ 3 levels "Driver & Passenger",...: 3
## $ DriveTrain        : Factor w/ 3 levels "4WD","Front",...: 2 2 2 2
## $ Cylinders         : Factor w/ 6 levels "3","4","5","6",...: 2 4 4
## $ EngineSize        : num   1.8 3.2 2.8 2.8 3.5 2.2 3.8 5.7 3.8 4.9
## $ Horsepower        : int   140 200 172 172 208 110 170 180 170 200
## $ RPM               : int   6300 5500 5500 5500 5700 5200 4800 4000
## $ Rev.per.mile       : int   2890 2335 2280 2535 2545 2565 1570 1320
## $ Man.trans.avail   : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 1 1
## $ Fuel.tank.capacity: num   13.2 18 16.9 21.1 21.1 16.4 18 23 18.8 1

```

```

8 ...
## $ Passengers      : int   5 5 5 6 4 6 6 6 5 6 ...
## $ Length          : int   177 195 180 193 186 189 200 216 198 206
...
## $ Wheelbase       : int   102 115 102 106 109 105 111 116 108 114
...
## $ Width           : int    68 71 67 70 69 69 74 78 73 73 ...
## $ Turn.circle     : int    37 38 37 37 39 41 42 45 41 43 ...
## $ Rear.seat.room  : num    26.5 30 28 31 27 28 30.5 30.5 26.5 35 ...
## $ Luggage.room    : int    11 15 14 17 13 16 17 21 14 18 ...
## $ Weight          : int   2705 3560 3375 3405 3640 2880 3470 4105
3495 3620 ...
## $ Origin          : Factor w/ 2 levels "USA","non-USA": 2 2 2 2 2
1 1 1 1 1 ...
## $ Make            : Factor w/ 93 levels "Acura Integra",...: 1 2 4
3 5 6 7 9 8 10 ...

```

# Rear.seat.room 等欄位有NA

summary(df)

| ##              | Manufacturer  | Model         | Type                 | Min.Price     | Price     |
|-----------------|---------------|---------------|----------------------|---------------|-----------|
| ## Chevrolet: 8 | 100           | : 1           | Compact:16           | Min. : 6.70   | Min. :    |
| 7.40            |               |               |                      |               |           |
| ## Ford : 8     | 190E          | : 1           | Large :11            | 1st Qu.:10.80 | 1st Qu.:1 |
| 2.20            |               |               |                      |               |           |
| ## Dodge : 6    | 240           | : 1           | Midsize:22           | Median :14.70 | Median :1 |
| 7.70            |               |               |                      |               |           |
| ## Mazda : 5    | 300E          | : 1           | Small :21            | Mean :17.13   | Mean :1   |
| 9.51            |               |               |                      |               |           |
| ## Pontiac : 5  | 323           | : 1           | Sporty :14           | 3rd Qu.:20.30 | 3rd Qu.:2 |
| 3.30            |               |               |                      |               |           |
| ## Buick : 4    | 535i          | : 1           | Van : 9              | Max. :45.40   | Max. :6   |
| 1.90            |               |               |                      |               |           |
| ## (Other) :57  | (Other):87    |               |                      |               |           |
| ## Max.Price    | MPG.city      | MPG.highway   |                      |               | AirBag    |
| s               |               |               |                      |               |           |
| ## Min. : 7.9   | Min. :15.00   | Min. :20.00   | Driver & Passenger:1 |               |           |
| 6               |               |               |                      |               |           |
| ## 1st Qu.:14.7 | 1st Qu.:18.00 | 1st Qu.:26.00 | Driver only          | :4            |           |
| 3               |               |               |                      |               |           |
| ## Median :19.6 | Median :21.00 | Median :28.00 | None                 | :3            |           |
| 4               |               |               |                      |               |           |
| ## Mean :21.9   | Mean :22.37   | Mean :29.09   |                      |               |           |
|                 |               |               |                      |               |           |
| ## 3rd Qu.:25.3 | 3rd Qu.:25.00 | 3rd Qu.:31.00 |                      |               |           |
|                 |               |               |                      |               |           |
| ## Max. :80.0   | Max. :46.00   | Max. :50.00   |                      |               |           |



```

##
## DriveTrain Cylinders EngineSize Horsepower RPM
## 4WD :10 3 : 3 Min. :1.000 Min. : 55.0 Min. :3800
## Front:67 4 :49 1st Qu.:1.800 1st Qu.:103.0 1st Qu.:4800
## Rear :16 5 : 2 Median :2.400 Median :140.0 Median :5200
## 6 :31 Mean :2.668 Mean :143.8 Mean :5281
## 8 : 7 3rd Qu.:3.300 3rd Qu.:170.0 3rd Qu.:5750
## rotary: 1 Max. :5.700 Max. :300.0 Max. :6500
##
## Rev.per.mile Man.trans.avail Fuel.tank.capacity Passengers
## Min. :1320 No :32 Min. : 9.20 Min. :2.000
## 1st Qu.:1985 Yes:61 1st Qu.:14.50 1st Qu.:4.000
## Median :2340 Median :16.40 Median :5.000
## Mean :2332 Mean :16.66 Mean :5.086
## 3rd Qu.:2565 3rd Qu.:18.80 3rd Qu.:6.000
## Max. :3755 Max. :27.00 Max. :8.000
##
## Length Wheelbase Width Turn.circle
## Min. :141.0 Min. : 90.0 Min. :60.00 Min. :32.00
## 1st Qu.:174.0 1st Qu.: 98.0 1st Qu.:67.00 1st Qu.:37.00
## Median :183.0 Median :103.0 Median :69.00 Median :39.00
## Mean :183.2 Mean :103.9 Mean :69.38 Mean :38.96
## 3rd Qu.:192.0 3rd Qu.:110.0 3rd Qu.:72.00 3rd Qu.:41.00
## Max. :219.0 Max. :119.0 Max. :78.00 Max. :45.00
##
## Rear.seat.room Luggage.room Weight Origin
## Make
## Min. :19.00 Min. : 6.00 Min. :1695 USA :48 Acura Integra: 1
## 1st Qu.:26.00 1st Qu.:12.00 1st Qu.:2620 non-USA:45 Acura Legend : 1
## Median :27.50 Median :14.00 Median :3040 Audi 100 : 1
## Mean :27.83 Mean :13.89 Mean :3073 Audi 90 : 1
## 3rd Qu.:30.00 3rd Qu.:15.00 3rd Qu.:3525 BMW 535i : 1
## Max. :36.00 Max. :22.00 Max. :4105 Buick Century: 1

```

```

entury: 1
## NAs :2 NAs :11 (Other)
:87

# 查看遺漏值
is.na(df[1:3, 1:5])

## Manufacturer Model Type Min.Price Price
## 1 FALSE FALSE FALSE FALSE FALSE
## 2 FALSE FALSE FALSE FALSE FALSE
## 3 FALSE FALSE FALSE FALSE FALSE

# 每欄遺漏值數量, Rear.seat.room, Luggage.room 有遺漏值
colSums(is.na(df))

## Manufacturer Model Type Mi
n.Price
## 0 0 0
## Price Max.Price MPG.city MPG.
highway
## 0 0 0
## AirBags DriveTrain Cylinders Eng
ineSize
## 0 0 0
## Horsepower RPM Rev.per.mile Man.tran
s.avail
## 0 0 0
## Fuel.tank.capacity Passengers Length Wh
eelbase
## 0 0 0
## Width Turn.circle Rear.seat.room Lugga
ge.room
## 0 0 2
11
## Weight Origin Make
## 0 0 0

# 每列遺漏值數量
rowSums(is.na(df))

## 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
24 25 26
## 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 2 0 0 0 0
0 0 1
## 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49
50 51 52

```

```
## 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
## 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75
76 77 78
## 0 0 0 1 2 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0
0 0 0
## 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93
## 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0
```

# 找出含 NA 的資料列

```
df[!complete.cases(df), ]
```

# 刪除遺漏值, df 資料沒有改變, 只是顯示執行結果

# 如果希望

```
na.omit(df)
```

## 填補遺漏值

### 方法一：平均數

```
df$Luggage.room[is.na(df$Luggage.room)] <- mean(df$Luggage.room, na.rm
= TRUE)
```

colSums(is.na(df)) # Luggage.room 欄位已經沒有遺漏值

```
##      Manufacturer      Model      Type      Mi
n.Price
##           0           0           0
0
##      Price      Max.Price      MPG.city      MPG.
highway
##           0           0           0
0
##      AirBags      DriveTrain      Cylinders      Eng
ineSize
##           0           0           0
0
##      Horsepower      RPM      Rev.per.mile      Man.tran
s.avail
##           0           0           0
0
## Fuel.tank.capacity      Passengers      Length      Wh
eelbase
##           0           0           0
0
##      Width      Turn.circle      Rear.seat.room      Lugga
ge.room
##           0           0           2
0
##      Weight      Origin      Make
##           0           0           0
```

方法二：中位數

```
df$Rear.seat.room <- median(df$Rear.seat.room, na.rm = TRUE)
sum(is.na(df[, 'Rear.seat.room']))

## [1] 0
```

方法三：固定值

```
# replace_na {tidyr}
```

## 2.重複值(Duplicate Records)

範例：

Bob 有重複。

| ID | Name  | Gender | Math | English |
|----|-------|--------|------|---------|
| 1  | ALAN  | F      | 95   | 80      |
| 2  | BOB   | M      | 70   | 85      |
| 2  | BOB   | M      | 70   | 85      |
| 4  | DAVID | M      | 90   | 77      |

```
df <- data.frame(
  ID      = c(1, 2, 2, 4),
  Name    = c("ALAN", "BOB", "BOB", "DAVID"),
  Gender  = c("F", "M", "M", "M"),
  Math    = c(95, 70, 70, 90),
  English = c(80, 85, 85, 77)
)
df
```

找出重複資料列

```
df[duplicated(df), ]
```

```
## [1] FALSE FALSE TRUE FALSE
```

查看重複資料

```
df[duplicated(df), ]
```

刪除重複資料列

```
df[!duplicated(df), ]
```

刪除重複資料列 {dplyr}

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

distinct(df)
```

### 3.資料型態錯誤(Incorrect Data Types)

#### 常見資料型態錯誤

| 問題                 | 常見原因          | 解決方式           |
|--------------------|---------------|----------------|
| 數值變成文字 (character) | Excel 混入文字、單位 | as.numeric()   |
| 日期變成文字             | CSV 匯入        | as.Date()      |
| 類別變成文字             | 性別、縣市         | factor()       |
| 一欄存多個資訊            | “男,25”        | separate()     |
| 多欄代表同一變數           | Jan、Feb、Mar   | pivot_longer() |
| 一欄混合不同型態           | “100kg”       | 去除單位再轉型        |
| 缺失值表示方式不一致         | “,”-“,”NA”    | 統一成 NA         |
| 重複資料               | 匯入重覆          | distinct()     |
| 欄名不一致              | Age、AGE、age   | rename()       |

- 字串轉換為數值: `df$Column <- as.numeric(df$Column)`
- 數值轉換為字串: `df$Column <- as.character(df$Column)`
- 字串轉換為日期: `df$Column <- as.Date(df$Column)`

#### 字串轉換為日期

```
tmp <- as.Date("31/08/2026", format="%d/%m/%Y")
print(tmp)

## [1] "2026-08-31"

class(tmp)

## [1] "Date"
```

#### 字串轉換為因子 (Factor)

```
# df$Column <- as.factor(df$Column)
```

因子轉換字串

```
# df$Column <- as.character(df$Column)
```

## 🌸 2.2 dplyr :

### filter/select/mutate/summarise/group\_by

#### 管線運算子(%>%)

- magrittr 套件中主要的功能是提供 %>% 管線運算子(Pipe Operator)
- 管線運算子使用 ‘|>’ 或 ‘%>%’ 是一種將前一個指令或函數的輸出結果 (output), 直接傳遞給下一個函數作為輸入的工具(input), 即做為右側函數的第一個參數輸入.
- 它能让程式碼由左至右、由上而下順暢閱讀, 提升可讀性避免多層巢狀括號的混亂.
- 管線運算子能減少暫存變數, 不需要為每一個中間步驟宣告額外的變數.
- RStudio Global Options Tools Code Editing: [Use native pipe operator,|> (requires R 4.1+)] 打勾
- CRAN magrittr 套件: <https://cran.r-project.org/web/packages/magrittr/index.html>
- magrittr 套件簡介: <https://cran.r-project.org/web/packages/magrittr/vignettes/magrittr.html>
- magrittr 套件範例

```
# 1 加侖 = 3.785 公升
# 1 公里 = 0.621371192 英里
# 原資料: miles per gallon
# 新資料: kilometers per liter
# 新資料 = miles per gallon*((1/0.621371192)/3.785)=0.42519

library(magrittr)

car_data <-
  mtcars %>%
  subset(hp > 100) %>%
  aggregate(. ~ cyl, data = ., FUN = . %>% mean %>% round(2)) %>%
  transform(kpl = mpg %>% multiply_by(0.4251)) %>%
  print
```

```
##   cyl   mpg   disp  hp drat   wt  qsec    vs  am gear carb
kpl
## 1    4 25.90 108.05 111.00 3.94 2.15 17.75 1.00 1.00 4.50 2.00 11.010
090
## 2    6 19.74 183.31 122.29 3.59 3.12 17.98 0.57 0.43 3.86 3.43  8.391
474
## 3    8 15.10 353.10 209.21 3.23 4.00 16.77 0.00 0.14 3.29 3.50  6.419
010
```

## 認識 dplyr 套件

- r cran dplyr: <https://cran.r-project.org/web/packages/dplyr/index.html>
- dplyr: A Grammar of Data Manipulation 資料操作文法
- dplyr = data frame + plyr
- plyr 發音 plier (鉗子), plyr 是 dplyr 的前個套件版本, 作者推薦使用 dplyr 套件
- plyr: <https://www.slideshare.net/hadley/plyr-one-data-analytic-strategy>

## flights 2013 年 NYC 機場航班資料, 33 萬筆資料

```
library(dplyr)
library(nycflights13)

data(flights) # tibble: 336776*19
print(flights)

## # A tibble: 336,776 × 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sche
d_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>
##   <int>
## 1  2013     1     1     517             515           2     830
819
## 2  2013     1     1     533             529           4     850
830
## 3  2013     1     1     542             540           2     923
850
## 4  2013     1     1     544             545          -1    1004
1022
## 5  2013     1     1     554             600          -6     812
837
## 6  2013     1     1     554             558          -4     740
728
## 7  2013     1     1     555             600          -5     913
854
## 8  2013     1     1     557             600          -3     709
723
## 9  2013     1     1     557             600          -3     838
846
## 10 2013     1     1     558             600          -2     753
```

```

745
## # i 336,766 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distanc
## #   e <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>

class(flights) # "tbl_df" "tbl" "data.frame"

## [1] "tbl_df"      "tbl"        "data.frame"

```

- flights 資料集變數說明

| 變數             | 資料型態      | 中文說明         | 範例                   |
|----------------|-----------|--------------|----------------------|
| year           | integer   | 年份           | 2013                 |
| month          | integer   | 月份           | 1~12                 |
| day            | integer   | 日期           | 1~31                 |
| dep_time       | integer   | 實際起飛時間(HHMM) | 517(表示 05:17), NA 取消 |
| sched_dep_time | integer   | 表定起飛時間(HHMM) | 515                  |
| dep_delay      | double    | 起飛延誤(分鐘)     | 2、-5、60              |
| arr_time       | integer   | 實際抵達時間(HHMM) | 830                  |
| sched_arr_time | integer   | 表定抵達時間(HHMM) | 819                  |
| arr_delay      | double    | 抵達延誤(分鐘)     | 11、-20               |
| carrier        | character | 航空公司代碼       | AA、DL                |
| flight         | integer   | 航班編號         | 1545                 |
| tailnum        | character | 飛機註冊編號       | N14228               |
| origin         | character | 起飛機場         | JFK                  |
| dest           | character | 目的地機場        | LAX                  |
| air_time       | double    | 飛行時間(分鐘)     | 150                  |
| distance       | double    | 飛行距離(英里)     | 950                  |
| hour           | double    | 表定起飛小時       | 5                    |
| min            | double    | 表定起飛分鐘       | 15                   |
| time_hour      | POSIXct   | 日期時間(UTC)    | 2013-01-01 05:00     |

- dep\_delay:

- dep\_delay 起飛延誤 = 實際起飛 - 表定起飛
- -10: 提早 10 分鐘
- 0: 準時



- 15: 延誤 15 分鐘
- 120: 延誤 2 小時
- arr\_delay:
  - arr\_delay 抵達延誤 = 實際抵達 - 表定抵達, 抵達延誤可用於分析航班績效重要的變數
  - 11: 晚 11 分鐘抵達
  - -20: 提早 20 分鐘抵達
- 資料檢視函數, 快速查看資料框 (data.frame 或 tibble) 的結構.

```
glimpse(flights)
## Rows: 336,776
## Columns: 19
## $ year          <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 201
3, 2013, 2...
## $ month         <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1...
## $ day           <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1...
## $ dep_time      <int> 517, 533, 542, 544, 554, 554, 555, 557, 557,
558, 558, ...
## $ sched_dep_time <int> 515, 529, 540, 545, 600, 558, 600, 600, 600,
600, 600, ...
## $ dep_delay     <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2,
-2, -2, -1...
## $ arr_time      <int> 830, 850, 923, 1004, 812, 740, 913, 709, 838,
753, 849,...
## $ sched_arr_time <int> 819, 830, 850, 1022, 837, 728, 854, 723, 846,
745, 851,...
## $ arr_delay     <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2,
-3, 7, -1...
## $ carrier       <chr> "UA", "UA", "AA", "B6", "DL", "UA", "B6", "EV
", "B6", "...
## $ flight        <int> 1545, 1714, 1141, 725, 461, 1696, 507, 5708,
79, 301, 4...
## $ tailnum       <chr> "N14228", "N24211", "N619AA", "N804JB", "N668
DN", "N394...
## $ origin        <chr> "EWR", "LGA", "JFK", "JFK", "LGA", "EWR", "EW
R", "LGA",...
## $ dest          <chr> "IAH", "IAH", "MIA", "BQN", "ATL", "ORD", "FL
L", "IAD",...
## $ air_time      <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 1
38, 149, 1...
## $ distance      <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229,
```

```

944, 733, ...
## $ hour          <dbl> 5, 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 6,
5, 6, 6, 6...
## $ minute        <dbl> 15, 29, 40, 45, 0, 58, 0, 0, 0, 0, 0, 0, 0, 0,
0, 59, 0...
## $ time_hour     <dtm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 20
13-01-01 0...

```

- 資料結構

```

str(flights)

## tibble [336,776 × 19] (S3: tbl_df/tbl/data.frame)
## $ year          : int [1:336776] 2013 2013 2013 2013 2013 2013 2013 2013
2013 2013 2013 ...
## $ month         : int [1:336776] 1 1 1 1 1 1 1 1 1 1 ...
## $ day           : int [1:336776] 1 1 1 1 1 1 1 1 1 1 ...
## $ dep_time      : int [1:336776] 517 533 542 544 554 554 555 557 55
7 558 ...
## $ sched_dep_time: int [1:336776] 515 529 540 545 600 558 600 600 60
0 600 ...
## $ dep_delay     : num [1:336776] 2 4 2 -1 -6 -4 -5 -3 -3 -2 ...
## $ arr_time      : int [1:336776] 830 850 923 1004 812 740 913 709 8
38 753 ...
## $ sched_arr_time: int [1:336776] 819 830 850 1022 837 728 854 723 8
46 745 ...
## $ arr_delay     : num [1:336776] 11 20 33 -18 -25 12 19 -14 -8 8 ...
## $ carrier       : chr [1:336776] "UA" "UA" "AA" "B6" ...
## $ flight        : int [1:336776] 1545 1714 1141 725 461 1696 507 57
08 79 301 ...
## $ tailnum       : chr [1:336776] "N14228" "N24211" "N619AA" "N804JB
" ...
## $ origin        : chr [1:336776] "EWR" "LGA" "JFK" "JFK" ...
## $ dest          : chr [1:336776] "IAH" "IAH" "MIA" "BQN" ...
## $ air_time      : num [1:336776] 227 227 160 183 116 150 158 53 140
138 ...
## $ distance      : num [1:336776] 1400 1416 1089 1576 762 ...
## $ hour          : num [1:336776] 5 5 5 5 6 5 6 6 6 6 ...
## $ minute        : num [1:336776] 15 29 40 45 0 58 0 0 0 0 ...
## $ time_hour     : POSIXct[1:336776], format: "2013-01-01 05:00:00"
"2013-01-01 05:00:00" ...

```

- 資料摘要

```

summary(flights) # 有NA

##      year      month      day      dep_time      sche
d_dep_time
## Min.   :2013   Min.    : 1.000   Min.    : 1.00   Min.    : 1   Min.
: 106
## 1st Qu.:2013   1st Qu.: 4.000   1st Qu.: 8.00   1st Qu.: 907   1st
Qu.: 906

```

```

## Median :2013      Median : 7.000      Median :16.00      Median :1401      Medi
an :1359
## Mean      :2013      Mean      : 6.549      Mean      :15.71      Mean      :1349      Mean
:1344
## 3rd Qu.:2013      3rd Qu.:10.000      3rd Qu.:23.00      3rd Qu.:1744      3rd
Qu.:1729
## Max.      :2013      Max.      :12.000      Max.      :31.00      Max.      :2400      Max.
:2359
##
##                                     NAs      :8255

##      dep_delay      arr_time      sched_arr_time      arr_delay
## Min.      : -43.00      Min.      : 1      Min.      : 1      Min.      : -86.000
## 1st Qu.: -5.00      1st Qu.:1104      1st Qu.:1124      1st Qu.: -17.000
## Median : -2.00      Median :1535      Median :1556      Median : -5.000
## Mean      : 12.64      Mean      :1502      Mean      :1536      Mean      : 6.895
## 3rd Qu.: 11.00      3rd Qu.:1940      3rd Qu.:1945      3rd Qu.: 14.000
## Max.      :1301.00      Max.      :2400      Max.      :2359      Max.      :1272.000
## NAs      :8255      NAs      :8713      NAs      :9430
##      carrier      flight      tailnum      origin

## Length      :336776      Min.      : 1      Length      :336776      Length      :3367
76
## N.unique : 16      1st Qu.: 553      N.unique : 4043      N.unique :
3
## N.blank : 0      Median :1496      N.blank : 0      N.blank :
0
## Min.nchar: 2      Mean      :1972      Min.nchar: 5      Min.nchar:
3
## Max.nchar: 2      3rd Qu.:3465      Max.nchar: 6      Max.nchar:
3
##                                     Max.      :8500      NAs      : 2512

##

##      dest      air_time      distance      hour
## Length      :336776      Min.      : 20.0      Min.      : 17      Min.      : 1.00
## N.unique : 105      1st Qu.: 82.0      1st Qu.: 502      1st Qu.: 9.00
## N.blank : 0      Median :129.0      Median : 872      Median :13.00
## Min.nchar: 3      Mean      :150.7      Mean      :1040      Mean      :13.18
## Max.nchar: 3      3rd Qu.:192.0      3rd Qu.:1389      3rd Qu.:17.00
##                                     Max.      :695.0      Max.      :4983      Max.      :23.00
##                                     NAs      :9430
##      minute      time_hour
## Min.      : 0.00      Min.      :2013-01-01 05:00:00
## 1st Qu.: 8.00      1st Qu.:2013-04-04 13:00:00
## Median :29.00      Median :2013-07-03 10:00:00
## Mean      :26.23      Mean      :2013-07-03 05:22:54
## 3rd Qu.:44.00      3rd Qu.:2013-10-01 07:00:00

```

```
## Max. :59.00 Max. :2013-12-31 23:00:00
##
```

- 如何轉換為 `tbl_df`, 使用 `tibble::as.tbl`

```
mytbl <- tibble::as_tibble(iris)
class(mytbl)
```

```
## [1] "tbl_df"      "tbl"        "data.frame"
```

- 前 6 筆資料

```
head(flights)
```

- 後 6 筆資料

*# 注意: 資料最後日期為 12 月 31 日, 但資料排序不是依照月, 日排序, 最後一筆是 9 月 30 日*

```
tail(flights)
```

## dplyr 套件重要函數

| 函數                     | 說明           |
|------------------------|--------------|
| <code>filter</code>    | 條件式篩選資料      |
| <code>slice</code>     | 列的指標篩選資料     |
| <code>arrange</code>   | 排序           |
| <code>select</code>    | 選取行/更改欄位名稱   |
| <code>rename</code>    | 選取所有行/更改欄位名稱 |
| <code>distinct</code>  | 選取不重覆資料      |
| <code>mutate</code>    | 新增欄位, 保留原資料  |
| <code>transmute</code> | 新增欄位, 不保留原資料 |
| <code>summarise</code> | 群組計算         |

### filter 條件式篩選資料

```
filter(flights, month == 1, day == 1)
```

`flights[flights$month == 1 & flights$day == 1, ]` *# 基本指令, same as filter*

```
filter(flights, month == 1 | month == 2) # AND 條件篩選, 同理 OR 使用 |
```

### slice 列的指標篩選資料

*# 第 1 筆*

```
slice(flights, 1)
```

*# 最後一筆*

```
slice(flights, n())
```

```
# 第123456 筆至最後一筆  
slice(flights, 123456:n())
```

arrange 排序

```
# 依照年,月,日遞增排序  
arrange(flights, year, month, day)  
  
# 依照延誤時間遞減排序  
arrange(flights, desc(arr_delay))
```

select 選取行

```
# 選取 3 行資料  
select(flights, year, month, day)
```

select 選取行,使用冒號(:)表示連續範圍

```
select(flights, year:dep_delay)
```

select 選取行,使用負號(-)表示刪除

```
select(flights, -(year:day))
```

select 選取行並且更改欄位名稱

```
# select 只選取 tailnum 1 行資料, 更名為 tail_num  
select(flights, tail_num = tailnum)
```

select 選取行,使用 starts\_with 等參數

```
iris %>% select(starts_with("Sepal"))
```

- starts\_with(): Starts with a prefix.
- ends\_with(): Ends with a suffix.
- contains(): Contains a literal string.
- matches(): Matches a regular expression.
- num\_range(): Matches a numerical range like 1:100.

rename 更改欄位名稱

```
rename(flights, ActualDepatureTime = dep_time)
```

distinct 選取不重覆資料

```
set.seed(168)
```

```
df <- data.frame(  
  x = sample(10, 100, replace = TRUE), # rep = replace
```

```

y = sample(10, 100, rep = TRUE)
) # 100*2

head(df)

# 不重複列
distinct(df)

# nrow 列數
nrow(distinct(df))

## [1] 63

nrow(distinct(df, x, y))

## [1] 63

distinct(df, x) # 與下列結果相同 unique(df$x)

distinct(df, y) # 與下列結果相同 unique(df$y)

```

## mutate 新增欄位

```

# mutate 新增欄位, 保留原資料
print(mutate(mtcars, displ_l = disp / 61.0237))

##           mpg  cyl  disp  hp drat   wt  qsec vs  am gear c
arb
## Mazda RX4           21.0    6 160.0 110 3.90 2.620 16.46  0  1    4
4
## Mazda RX4 Wag       21.0    6 160.0 110 3.90 2.875 17.02  0  1    4
4
## Datsun 710           22.8    4 108.0  93 3.85 2.320 18.61  1  1    4
1
## Hornet 4 Drive       21.4    6 258.0 110 3.08 3.215 19.44  1  0    3
1
## Hornet Sportabout    18.7    8 360.0 175 3.15 3.440 17.02  0  0    3
2
## Valiant              18.1    6 225.0 105 2.76 3.460 20.22  1  0    3
1
## Duster 360           14.3    8 360.0 245 3.21 3.570 15.84  0  0    3
4
## Merc 240D            24.4    4 146.7  62 3.69 3.190 20.00  1  0    4
2
## Merc 230             22.8    4 140.8  95 3.92 3.150 22.90  1  0    4
2
## Merc 280             19.2    6 167.6 123 3.92 3.440 18.30  1  0    4
4
## Merc 280C            17.8    6 167.6 123 3.92 3.440 18.90  1  0    4
4
## Merc 450SE           16.4    8 275.8 180 3.07 4.070 17.40  0  0    3

```

|    |                     |      |   |       |     |      |       |       |   |   |   |
|----|---------------------|------|---|-------|-----|------|-------|-------|---|---|---|
| 3  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Merc 450SL          | 17.3 | 8 | 275.8 | 180 | 3.07 | 3.730 | 17.60 | 0 | 0 | 3 |
| 3  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Merc 450SLC         | 15.2 | 8 | 275.8 | 180 | 3.07 | 3.780 | 18.00 | 0 | 0 | 3 |
| 3  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Cadillac Fleetwood  | 10.4 | 8 | 472.0 | 205 | 2.93 | 5.250 | 17.98 | 0 | 0 | 3 |
| 4  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Lincoln Continental | 10.4 | 8 | 460.0 | 215 | 3.00 | 5.424 | 17.82 | 0 | 0 | 3 |
| 4  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Chrysler Imperial   | 14.7 | 8 | 440.0 | 230 | 3.23 | 5.345 | 17.42 | 0 | 0 | 3 |
| 4  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Fiat 128            | 32.4 | 4 | 78.7  | 66  | 4.08 | 2.200 | 19.47 | 1 | 1 | 4 |
| 1  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Honda Civic         | 30.4 | 4 | 75.7  | 52  | 4.93 | 1.615 | 18.52 | 1 | 1 | 4 |
| 2  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Toyota Corolla      | 33.9 | 4 | 71.1  | 65  | 4.22 | 1.835 | 19.90 | 1 | 1 | 4 |
| 1  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Toyota Corona       | 21.5 | 4 | 120.1 | 97  | 3.70 | 2.465 | 20.01 | 1 | 0 | 3 |
| 1  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Dodge Challenger    | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 | 0 | 0 | 3 |
| 2  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | AMC Javelin         | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 | 0 | 0 | 3 |
| 2  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Camaro Z28          | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 | 0 | 0 | 3 |
| 4  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Pontiac Firebird    | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 | 0 | 0 | 3 |
| 2  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Fiat X1-9           | 27.3 | 4 | 79.0  | 66  | 4.08 | 1.935 | 18.90 | 1 | 1 | 4 |
| 1  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Porsche 914-2       | 26.0 | 4 | 120.3 | 91  | 4.43 | 2.140 | 16.70 | 0 | 1 | 5 |
| 2  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Lotus Europa        | 30.4 | 4 | 95.1  | 113 | 3.77 | 1.513 | 16.90 | 1 | 1 | 5 |
| 2  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Ford Pantera L      | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 | 0 | 1 | 5 |
| 4  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Ferrari Dino        | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 | 0 | 1 | 5 |
| 6  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Maserati Bora       | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 | 0 | 1 | 5 |
| 8  |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Volvo 142E          | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 | 1 | 1 | 4 |
| 2  |                     |      |   |       |     |      |       |       |   |   |   |
| ## |                     |      |   |       |     |      |       |       |   |   |   |
| ## |                     |      |   |       |     |      |       |       |   |   |   |
| ## | Mazda RX4           |      |   |       |     |      |       |       |   |   |   |
| ## | Mazda RX4 Wag       |      |   |       |     |      |       |       |   |   |   |
| ## | Datsun 710          |      |   |       |     |      |       |       |   |   |   |
| ## | Hornet 4 Drive      |      |   |       |     |      |       |       |   |   |   |
| ## | Hornet Sportabout   |      |   |       |     |      |       |       |   |   |   |
| ## | Valiant             |      |   |       |     |      |       |       |   |   |   |
| ## | Duster 360          |      |   |       |     |      |       |       |   |   |   |
| ## | Merc 240D           |      |   |       |     |      |       |       |   |   |   |

```
## Merc 230      2.307300
## Merc 280      2.746474
## Merc 280C     2.746474
## Merc 450SE    4.519556
## Merc 450SL    4.519556
## Merc 450SLC   4.519556
## Cadillac Fleetwood 7.734700
## Lincoln Continental 7.538055
## Chrysler Imperial 7.210313
## Fiat 128      1.289663
## Honda Civic   1.240502
## Toyota Corolla 1.165121
## Toyota Corona 1.968088
## Dodge Challenger 5.211090
## AMC Javelin   4.981671
## Camaro Z28    5.735477
## Pontiac Firebird 6.554830
## Fiat X1-9     1.294579
## Porsche 914-2 1.971365
## Lotus Europa  1.558411
## Ford Pantera L 5.751864
## Ferrari Dino   2.376126
## Maserati Bora  4.932510
## Volvo 142E    1.982836
```

## summarise - 計算加總

```
# 計算航班總數
flights %>%
  summarise(
    total_flights = n() # n() : 資料筆數
  )
```

## summarise - 平均起飛延誤

```
flights %>%
  summarise(
    mean_dep_delay = mean(dep_delay, na.rm = TRUE)
  )
```

## summarise - 多個統計量

```
flights %>%
  summarise(
    avg = mean(dep_delay, na.rm = TRUE),
    sd = sd(dep_delay, na.rm = TRUE),
    min = min(dep_delay, na.rm = TRUE),
    max = max(dep_delay, na.rm = TRUE)
  )
```



## group\_by + summarise

```
# 每家航空公司的平均延誤
flights %>%
  group_by(carrier) %>%
  summarise(
    avg_delay = mean(dep_delay, na.rm = TRUE)
  )

# 計算航空公司的班次總計
flights %>%
  group_by(carrier) %>%
  summarise(
    flights = n()
  )
```

## group\_by 多群組 + summarise

```
# 依航空公司與月份
flights %>%
  group_by(carrier, month) %>%
  summarise(
    avg_delay = mean(dep_delay, na.rm = TRUE)
  )

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by carrier and month.
## i Output is grouped by carrier.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(carrier, month))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

## 計算航空公司的平均延誤時間, 由大至小排序

```
flights %>%
  group_by(carrier) %>%
  summarise(
    avg_delay = mean(dep_delay, na.rm = TRUE)
  ) %>%
  arrange(desc(avg_delay))
```

## dplyr in R Cheat Sheet

<https://nyu-cdsc.github.io/learningr/assets/data-transformation.pdf>

## 熟悉開放資料集於 dplyr 應用

- 2025.07.12\_DATASETS\_WISHLIST
  - [https://github.com/rwepa/r\\_data\\_scientist/tree/main/r\\_training\\_advanced\\_ppt\\_codes\\_2025#20250712\\_datasets\\_wishlist](https://github.com/rwepa/r_data_scientist/tree/main/r_training_advanced_ppt_codes_2025#20250712_datasets_wishlist)

## 🌸 2.3 tidyr : pivot\_longer/pivot\_wider、分欄/合欄

### 長寬資料簡介

- tidyr 套件中的 pivot\_longer()（將寬表格轉為長表格）與 pivot\_wider()（將長表格轉為寬表格）是 R 語言中重塑資料（Reshaping Data）的核心函數。
- pivot\_longer 寬轉長: 當你的資料中欄位名稱包含資料變數（例如把年份 2021, 2022 放成欄位名）時, 非常適合使用 pivot\_longer.
- pivot\_longer 範例

```
library(tidyr)

##
## Attaching package: 'tidyr'

## The following object is masked from 'package:magrittr':
##
##      extract

library(dplyr)

# 建立寬表格 (Wide Format)
df_wide <- tibble(
  student = c("Alice", "Bob"),
  Math = c(90, 80),
  English = c(85, 95)
)

print(df_wide)

## # A tibble: 2 × 3
##   student Math English
##   <chr>   <dbl>   <dbl>
## 1 Alice     90     85
## 2 Bob      80     95

# 轉為長表格 (Long Format)
df_long <- df_wide %>%
  pivot_longer(
    cols = c(Math, English), # 要合併的欄位 (也可以寫 -student)
    names_to = "subject",    # 舊欄位名稱要放入的新欄位名稱
    values_to = "score"      # 舊欄位數值要放入的新欄位名稱
  )
```

```
print(df_long) # subject 欄位已經改為長資料
```

```
## # A tibble: 4 × 3
##   student subject score
##   <chr>    <chr>   <dbl>
## 1 Alice    Math      90
## 2 Alice    English   85
## 3 Bob      Math      80
## 4 Bob      English   95
```

- `pivot_wider` 長轉寬: 當你需要將資料做交叉表統計, 或是有些欄位的類別需要拆分成獨立欄位時, 使用 `pivot_wider`.
- `pivot_wider` 範例

```
# 轉為寬表格 (Wide Format)
```

```
df_recreated_wide <- df_long %>%
```

```
  pivot_wider(
    names_from = subject, # 哪一個欄位的值做為新欄位名稱
    values_from = score   # 哪一個欄位的值填入新欄位中
  )
```

```
print(df_recreated_wide)
```

```
## # A tibble: 2 × 3
##   student Math English
##   <chr>   <dbl>   <dbl>
## 1 Alice     90     85
## 2 Bob       80     95
```

- 寬表格的欄位名包含多個資訊 (例如 `Math_2021`, `Math_2022`)

```
# 包含科目與年份的複合欄位名
```

```
df_complex <- tibble(
  student = c("Alice", "Bob"),
  Math_2021 = c(90, 80),
  Math_2022 = c(92, 85),
  English_2021 = c(85, 95),
  English_2022 = c(88, 98)
)
```

```
print(df_complex)
```

```
## # A tibble: 2 × 5
##   student Math_2021 Math_2022 English_2021 English_2022
##   <chr>      <dbl>    <dbl>      <dbl>      <dbl>
## 1 Alice         90        92         85         88
## 2 Bob           80        85         95         98
```

```
# 使用 names_sep 或 names_pattern 一次拆分成兩個新欄位
```

```
df_complex_long <- df_complex %>%
```

```

pivot_longer(
  cols = -student,          # 負號表示除了此欄位的其他所有欄位
  names_to = c("subject", "year"), # 長資料的 2 個新欄位名稱
  names_sep = "_",          # 根據底線拆分
  values_to = "score"       # 長資料的值
)

print(df_complex_long)

## # A tibble: 8 × 4
##   student subject year  score
##   <chr>    <chr>   <chr> <dbl>
## 1 Alice    Math     2021    90
## 2 Alice    Math     2022    92
## 3 Alice    English 2021    85
## 4 Alice    English 2022    88
## 5 Bob      Math     2021    80
## 6 Bob      Math     2022    85
## 7 Bob      English 2021    95
## 8 Bob      English 2022    98

```

## tidyr + ggplot2 → 使用 tidyverse 整合套件

### # 1. 載入所需套件

```
library(tidyverse) # 包含 tidyr, ggplot2, dplyr, readr 等
```

```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
```

```
## ✓ forcats 1.0.1 ✓ purrr 1.2.2
```

```
## ✓ ggplot2 4.0.3 ✓ readr 2.2.0
```

```
## ✓ lubridate 1.9.5 ✓ stringr 1.6.0
```

```
## — Conflicts ————— tidyverse_conflicts() —
```

```
## ✗ tidyr::extract() masks magrittr::extract()
```

```
## ✗ dplyr::filter() masks stats::filter()
```

```
## ✗ dplyr::lag() masks stats::lag()
```

```
## ✗ purrr::set_names() masks magrittr::set_names()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

### # 2. 建立模擬資料 (寬表格：包含 Month, Product\_A, Product\_B, Product\_C)

```
set.seed(42)
```

```
df_sales_wide <- tibble(
  month = 1:12,
  Product_A = cumsum(rnorm(12, mean = 5, sd = 2)) + 100,
  Product_B = cumsum(rnorm(12, mean = 3, sd = 3)) + 80,
  Product_C = cumsum(rnorm(12, mean = 7, sd = 1.5)) + 60
)
```

```

print("原始寬表格 (df_sales_wide) : ")

## [1] "原始寬表格 (df_sales_wide) : "

print(head(df_sales_wide, 3))

## # A tibble: 3 × 4
##   month Product_A Product_B Product_C
##   <int>      <dbl>      <dbl>      <dbl>
## 1     1         108.        78.8        69.8
## 2     2         112.        81.0        76.2
## 3     3         117.        83.6        82.8

# 3. 使用 pivot_longer 轉為長表格
df_sales_long <- df_sales_wide %>%
  pivot_longer(
    cols = starts_with("Product_"), # 選擇要合併的欄位 (也可以寫 -month)
    names_to = "product_type",      # 舊欄位名稱轉入此新欄位
    values_to = "sales_amount"     # 舊欄位數值轉入此新欄位
  )

print("轉換後的長表格 (df_sales_long) : ")

## [1] "轉換後的長表格 (df_sales_long) : "

print(head(df_sales_long, 6))

## # A tibble: 6 × 3
##   month product_type sales_amount
##   <int> <chr>          <dbl>
## 1     1 Product_A        108.
## 2     1 Product_B        78.8
## 3     1 Product_C        69.8
## 4     2 Product_A        112.
## 5     2 Product_B        81.0
## 6     2 Product_C        76.2

# 4. 結合 ggplot2 繪製多指標折線圖
ggplot(df_sales_long, aes(x = month, y = sales_amount, color = product_
type)) +

  # 繪製折線與資料點
  geom_line(linewidth = 1) +
  geom_point(size = 2.5) +

  # 設定 X 軸刻度 (顯示 1~12 月)
  scale_x_continuous(breaks = 1:12) +

  # 設定顏色與圖例名稱

```

```

scale_color_brewer(
  palette = "Set1",
  name = "產品類別",
  labels = c("產品 A", "產品 B", "產品 C")
) +

# 標題與軸標籤
labs(
  title = "2025 年各產品月銷售額趨勢圖",
  subtitle = "使用 tidyr::pivot_longer 轉檔後繪製",
  x = "月份",
  y = "銷售額 (萬元)",
  caption = "資料來源：00 企業銷售部"
) +

# 套用簡潔主題與細節微調
theme_minimal(base_size = 12) + # 簡潔主題風格(無背景填色，無外框線，保留網格線)，基礎文字大小=12
theme(

  # 將標題置中對齊 (hjust = 0.5)，並加粗
  plot.title = element_text(hjust = 0.5, face = "bold", size = 16),

  # 副標題一起置中對齊
  plot.subtitle = element_text(hjust = 0.5, size = 12, color = "gray30"),

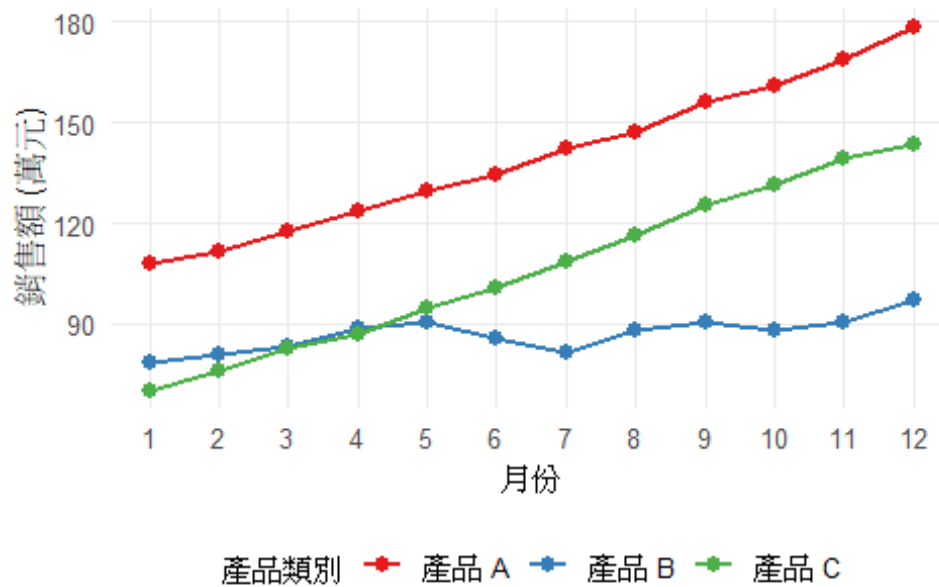
  # 將圖例移至下方
  legend.position = "bottom",

  # 移除次要網格線
  panel.grid.minor = element_blank()
)

```

## 2025 年各產品月銷售額趨勢圖

使用 `tidyr::pivot_longer` 轉檔後繪製



資料來源：OO企業銷售部

## 近十年國防預算結構分析

- 下載: <https://data.gov.tw/dataset/139911>
- 檔名: 近十年國防預算結構分析\_290973.csv

# 載入套件

```
library(readr)
```

```
library(dplyr)
```

# 匯入 CSV

```
df <- read_csv("近十年國防預算結構分析_290973.csv")
```

```
## Rows: 46 Columns: 8
```

```
## — Column specification —————
```

```
## Delimiter: ","
```

```
## chr (4): ItemName, Category, Author, URL
```

```
## dbl (4): SeqNo, Annual, Percentage, IssueDate
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

# 顯示資料集

```
df # 46*8
```

- 資料檢視函數,快速查看資料框 (data.frame 或 tibble) 的結構.

```
glimpse(df)
```

```
## Rows: 46
## Columns: 8
## $ SeqNo      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,
  16, 17, ...
## $ Annual     <dbl> 101, 101, 101, 101, 102, 102, 102, 102, 103, 103,
  103, 103,...
## $ ItemName   <chr> "近十年國防預算結構分析", "近十年國防預算結構分析", "
  近十年國防預算結構分析", "近十年國防預算結構分析",...
## $ Category   <chr> "軍事投資", "作業維持", "人員維持", "其他", "軍事投
  資", "作業維持", "人員維持", "其他",...
## $ Percentage <dbl> 27.26, 22.33, 49.00, 1.41, 25.98, 22.62, 49.92, 1.
  48, 27.62...
## $ Author     <chr> "國防部主計局", "國防部主計局", "國防部主計局", "國防
  部主計局", "國防部主計局", "國防部主計局",...
## $ IssueDate  <dbl> 20210521, 20210521, 20210521, 20210521, 20210521,
  20210521,...
## $ URL        <chr> "https://www.mnd.gov.tw/NewUpload/202105/12.%e8%b
  f%91%e5%8d...
```

- 資料結構

```
str(df)
```

```
## spc_tbl_ [46 × 8] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ SeqNo      : num [1:46] 1 2 3 4 5 6 7 8 9 10 ...
## $ Annual     : num [1:46] 101 101 101 101 102 102 102 102 103 103 ...
## $ ItemName   : chr [1:46] "近十年國防預算結構分析" "近十年國防預算結構分
  析" "近十年國防預算結構分析" "近十年國防預算結構分析" ...
## $ Category   : chr [1:46] "軍事投資" "作業維持" "人員維持" "其他" ...
## $ Percentage: num [1:46] 27.26 22.33 49 1.41 25.98 ...
## $ Author     : chr [1:46] "國防部主計局" "國防部主計局" "國防部主計局"
  "國防部主計局" ...
## $ IssueDate  : num [1:46] 20210521 20210521 20210521 20210521 202105
  21 ...
## $ URL        : chr [1:46] "https://www.mnd.gov.tw/NewUpload/202105/1
  2.%e8%bf%91%e5%8d%81%e5%b9%b4%e5%9c%8b%e9%98%b2%e9%a0%90%e7%ae%97%e7%b"
  | __truncated__ "https://www.mnd.gov.tw/NewUpload/202105/12.%e8%bf%91%e
  5%8d%81%e5%b9%b4%e5%9c%8b%e9%98%b2%e9%a0%90%e7%ae%97%e7%b"| __truncated
  __ "https://www.mnd.gov.tw/NewUpload/202105/12.%e8%bf%91%e5%8d%81%e5%b9%
  b4%e5%9c%8b%e9%98%b2%e9%a0%90%e7%ae%97%e7%b"| __truncated__ "https://ww
  w.mnd.gov.tw/NewUpload/202105/12.%e8%bf%91%e5%8d%81%e5%b9%b4%e5%9c%8b%e
  9%98%b2%e9%a0%90%e7%ae%97%e7%b"| __truncated__ ...
## - attr(*, "spec")=
## .. cols(
## ..   SeqNo = col_double(),
## ..   Annual = col_double(),
```



```
## .. ItemName = col_character(),
## .. Category = col_character(),
## .. Percentage = col_double(),
## .. Author = col_character(),
## .. IssueDate = col_double(),
## .. URL = col_character()
## .. )
## - attr(*, "problems")=<pointer: 0x000001833b88a8c0>
```

- 資料摘要

`summary(df)` # 有NA

```
##      SeqNo      Annual      ItemName      Category      Per
centage
## Min.   : 1.00   Min.   :101.0   Length   :46   Length   :46   Min.
: 1.41
## 1st Qu.:12.25   1st Qu.:103.2   N.unique : 1   N.unique : 4   1st Q
u.:22.95
## Median :23.50   Median :106.0   N.blank  : 0   N.blank  : 0   Media
n :27.24
## Mean   :23.50   Mean    :106.5   Min.nchar:11   Min.nchar: 2   Mean
:28.26
## 3rd Qu.:34.75   3rd Qu.:109.8   Max.nchar:11   Max.nchar: 4   3rd Q
u.:43.21
## Max.    :46.00   Max.     :113.0                      Max.
:49.92
##      Author      IssueDate      URL
## Length   :46   Min.    :20210521   Length   : 46
## N.unique : 1   1st Qu.:20210521   N.unique : 4
## N.blank  : 0   Median :20210521   N.blank  : 0
## Min.nchar: 6   Mean    :20215100   Min.nchar:153
## Max.nchar: 6   3rd Qu.:20210521   Max.nchar:181
##                      Max.    :20240531
```

# 前6筆

`head(df)`

# 後6筆

`tail(df)`

長資料(1 行軍事投資,作業維持,人員維持,其他)轉換為寬資料(4 行)

```
library(tidyr) # spread
library(dplyr)
```

方法 1 不使用管線操作

```
defense_long <- select(df, Annual:Percentage)
```

```
defense_long <- select(defense_long, -c(ItemName))

# spread 表示將長資料轉換為寬資料
defense_wide <- spread(defense_long, key=Category, value=Percentage)

defense_wide
```

## 方法 2 使用 %>%

```
defense_wide <- df %>%
  select(Annual:Percentage) %>%
  select(-c(ItemName)) %>%
  spread(key=Category, value=Percentage)

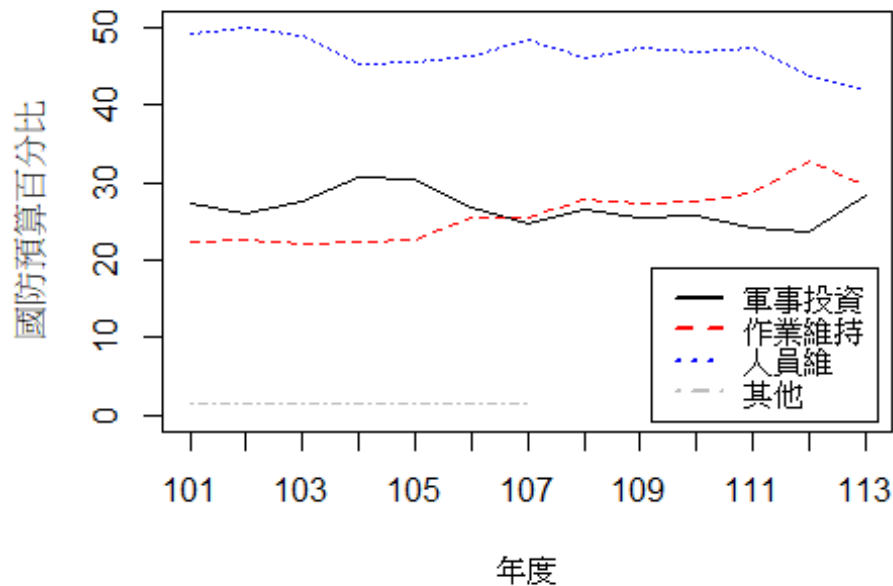
defense_wide
```

## 國防預算結構統計圖

```
plot(defense_wide$軍事投資, type="l",
     xlab="年度",
     ylab="國防預算百分比",
     main="101~113 年國防預算結構統計圖",
     axes=FALSE,
     cex=0.5,
     ylim=c(0, max(defense_wide[2:5], na.rm=TRUE)))
lines(defense_wide$作業維持, col="red", lty=2)
lines(defense_wide$人員維持, col="blue", lty=3)
lines(defense_wide$其他, col="grey", lty=4)

#ind <- round(seq(1, nrow(defense_wide), length.out=5))
axis(1, at=c(1:13), labels=defense_wide$Annual)
axis(2)
legend(x="bottomright",
      inset = c(0.02, 0.02),
      legend=c("軍事投資", "作業維持", "人員維", "其他"),
      col=c("black", "red", "blue", "grey"),
      lty=c(1:4),
      lwd=2,
      cex = 0.8)
box()
```

101~113年國防預算結構統計圖



```
# Legend(x 設定值= "xxx")
# topleft    top    topright
# left      center right
# bottomleft bottom bottomright
```

- 練習使用 `pivot_wider`, `pivot_longer` {tidyr} 進行長寬資料轉換

## 🌸 2.4 合併資料：join 家族與鍵值設計

### 合併簡介

- `tidyverse` 中的 `join` 是資料整合 (Data Joining) 的核心功能，主要由 `dplyr` 套件提供。它可以依據一個或多個共同欄位 (Key)，將兩個資料框 (`data.frame` 或 `tibble`) 合併，功能類似 SQL 的 JOIN。
- Join 的目的通常是「將右表 (第二個資料框) 的欄位加入左表 (第一個資料框)」。
- 共同欄位 (ID) 只是拿來配對，一般不會新增右表之共同欄位。
- `dplyr` 套件六種合併資料

| 函數 | 保留左表 | 保留右表 | 新增右表欄位 | 常見用途 |
|----|------|------|--------|------|
|----|------|------|--------|------|

| 函數                        | 保留左表    | 保留右表  | 新增右表欄位 | 常見用途       |
|---------------------------|---------|-------|--------|------------|
| <code>inner_join()</code> | 共同資料    | 共同資料  | ✓      | 交集分析       |
| <code>left_join()</code>  | ✓       | 只有符合者 | ✓      | 最常用，補充欄位   |
| <code>right_join()</code> | 只有符合者   | ✓     | ✓      | 保留右表       |
| <code>full_join()</code>  | ✓       | ✓     | ✓      | 完整整合資料     |
| <code>semi_join()</code>  | ✓（符合者）  | ✗     | ✗      | 篩選存在於右表的資料 |
| <code>anti_join()</code>  | ✓（未符合者） | ✗     | ✗      | 找遺漏、異常資料   |

- 註 1: `semi_join()` 的用途不是合併資料，而是依據右表是否存在符合條件的鍵值來篩選左表資料，因此只回傳左表原本的欄位。
- 註 2: `anti_join()` 的目的也是篩選，只留下左表中找不到配對的列，因此同樣不會加入右表的任何欄位。

## 範例 1 `left_join` - 左表新增右表的 1 個欄位

```
library(dplyr)

students <- tibble(
  ID = c(1,2,3),
  Name = c("Alan", "Bob", "Cathy")
)
print("student - 左表")

## [1] "student - 左表"

print(students)

## # A tibble: 3 × 2
##       ID Name
##   <dbl> <chr>
## 1     1 Alan
## 2     2 Bob
## 3     3 Cathy

scores <- tibble(
  ID = c(1,2,3),
  Math = c(90,80,95)
)
print("scores - 右表")

## [1] "scores - 右表"

print(scores)
```

```
## # A tibble: 3 × 2
##       ID  Math
##   <dbl> <dbl>
## 1     1    90
## 2     2    80
## 3     3    95

students %>%
  left_join(scores, by = "ID")
```

## 範例 2 left\_join - 左表新增右表的多個欄位

```
scores <- tibble(
  ID = c(1,2,3),
  Math = c(90,80,95),
  English = c(88,76,91),
  Science = c(92,81,94)
)

students %>%
  left_join(scores, by="ID")
```

## 範例 3 left\_join - 右表的欄位找不到

```
print(students)

## # A tibble: 3 × 2
##       ID Name
##   <dbl> <chr>
## 1     1  Alan
## 2     2   Bob
## 3     3  Cathy

scores <- tibble(
  ID = c(1,2),
  Math = c(90,80)
)
print("scores - 右表")

## [1] "scores - 右表"

print(scores)

## # A tibble: 2 × 2
##       ID  Math
##   <dbl> <dbl>
## 1     1    90
## 2     2    80

# 新增結果為 NA
students %>%
  left_join(scores)
```

```
## Joining with `by = join_by(ID)`
```

## 🌸 2.5 練習：完成一份可分析的乾淨資料集

參考 Open Data 等資料，完成 Tidy Data。

熟悉以下資料集之資料操作

1. AirPassengers : ts 時間序列
2. Titanic : table 表格
3. airquality : data.frame 資料框
4. iris : data.frame 資料框
5. mtcars : data.frame 資料框
6. penguins : data.frame 資料框

### 1. AirPassengers 空中旅客人數資料集

- 1949-1960 年航空旅客每月平均乘客數,單位:千人.
- 12 年\*12 個月=144 筆資料
- 時間序列是一組按照時間發生先後順序進行排列的資料序列.
- 通常一組時間序列的時間間隔為一固定值.
- [https://github.com/rwepa/r\\_data\\_scientist/blob/main/r\\_training\\_ppt\\_codes\\_2025/r\\_data\\_science\\_02\\_r\\_object.R#L652](https://github.com/rwepa/r_data_scientist/blob/main/r_training_ppt_codes_2025/r_data_science_02_r_object.R#L652)

```
?AirPassengers
```

```
## starting httpd help server ... done
```

```
data("AirPassengers")
```

```
AirPassengers
```

```
##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 112 118 132 129 121 135 148 148 136 119 104 118
## 1950 115 126 141 135 125 149 170 170 158 133 114 140
## 1951 145 150 178 163 172 178 199 199 184 162 146 166
## 1952 171 180 193 181 183 218 230 242 209 191 172 194
## 1953 196 196 236 235 229 243 264 272 237 211 180 201
## 1954 204 188 235 227 234 264 302 293 259 229 203 229
## 1955 242 233 267 269 270 315 364 347 312 274 237 278
## 1956 284 277 317 313 318 374 413 405 355 306 271 306
## 1957 315 301 356 348 355 422 465 467 404 347 305 336
## 1958 340 318 362 348 363 435 491 505 404 359 310 337
## 1959 360 342 406 396 420 472 548 559 463 407 362 405
## 1960 417 391 419 461 472 535 622 606 508 461 390 432
```

```
class(AirPassengers) # ts
```

```
## [1] "ts"

length(AirPassengers)

## [1] 144

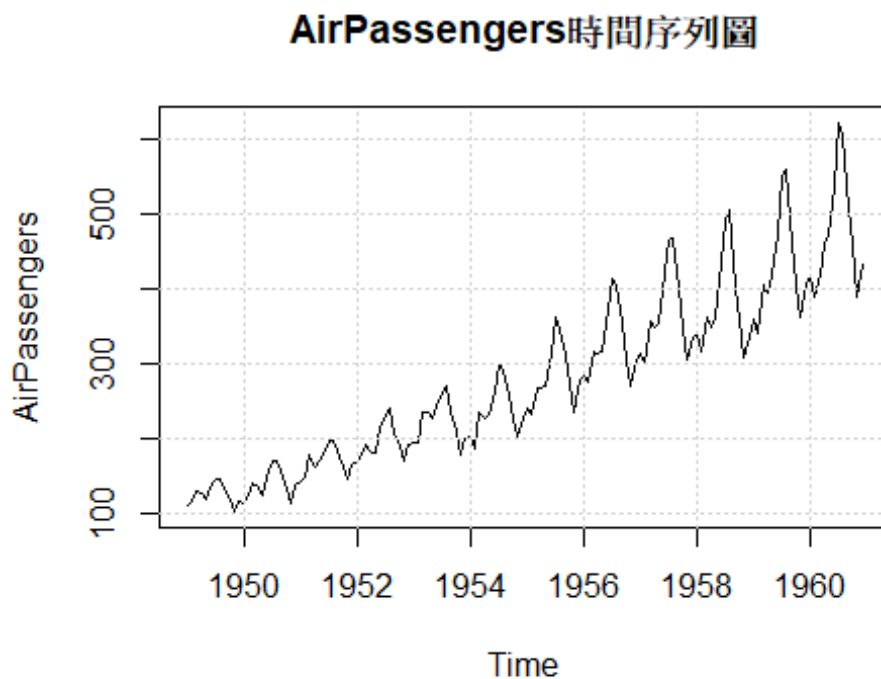
str(AirPassengers)

## Time-Series [1:144] from 1949 to 1961: 112 118 132 129 121 135 148
148 136 119 ...

summary(AirPassengers)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    104.0   180.0   265.5   280.3   360.5   622.0

# 繪製 ts 資料
plot(AirPassengers, main="AirPassengers 時間序列圖")
grid()
```



```
# 找出樣式 (Patterns)
```

## 2. Titanic 1912 年鐵達尼號資料集

```
# ?Titanic
```

```
data(Titanic)
```

Titanic # Titanic 不是 data.frame, 而是一個四維列聯表 (Contingency Table)

```
## , , Age = Child, Survived = No
```

```
##
```

```
##      Sex
```

```
## Class Male Female
```

```
## 1st      0      0
```

```
## 2nd      0      0
```

```
## 3rd     35     17
```

```
## Crew      0      0
```

```
##
```

```
## , , Age = Adult, Survived = No
```

```
##
```

```
##      Sex
```

```
## Class Male Female
```

```
## 1st    118      4
```

```
## 2nd    154     13
```

```
## 3rd    387     89
```

```
## Crew   670      3
```

```
##
```

```
## , , Age = Child, Survived = Yes
```

```
##
```

```
##      Sex
```

```
## Class Male Female
```

```
## 1st      5      1
```

```
## 2nd     11     13
```

```
## 3rd     13     14
```

```
## Crew      0      0
```

```
##
```

```
## , , Age = Adult, Survived = Yes
```

```
##
```

```
##      Sex
```

```
## Class Male Female
```

```
## 1st     57    140
```

```
## 2nd     14     80
```

```
## 3rd     75     76
```

```
## Crew   192     20
```

```
class(Titanic) # table
```

```
## [1] "table"
```

```
sum(Titanic) # 總人數
```

```
## [1] 2201
```

# 物件結構, 有4個維度(列: Class, 行: Sex, 第3維度: Age, 第4維度: Survived)

#  $4 \times 2 \times 2 \times 2 = 32$  種組合

```
str(Titanic)
```



```
## 'table' num [1:4, 1:2, 1:2, 1:2] 0 0 35 0 0 0 17 0 118 154 ...
## - attr(*, "dimnames")=List of 4
## ..$ Class : chr [1:4] "1st" "2nd" "3rd" "Crew"
## ..$ Sex : chr [1:2] "Male" "Female"
## ..$ Age : chr [1:2] "Child" "Adult"
## ..$ Survived: chr [1:2] "No" "Yes"
```

```
summary(Titanic)
```

```
## Number of cases in table: 2201
## Number of factors: 4
## Test for independence of all factors:
## Chisq = 1637.4, df = 25, p-value = 0
## Chi-squared approximation may be incorrect
```

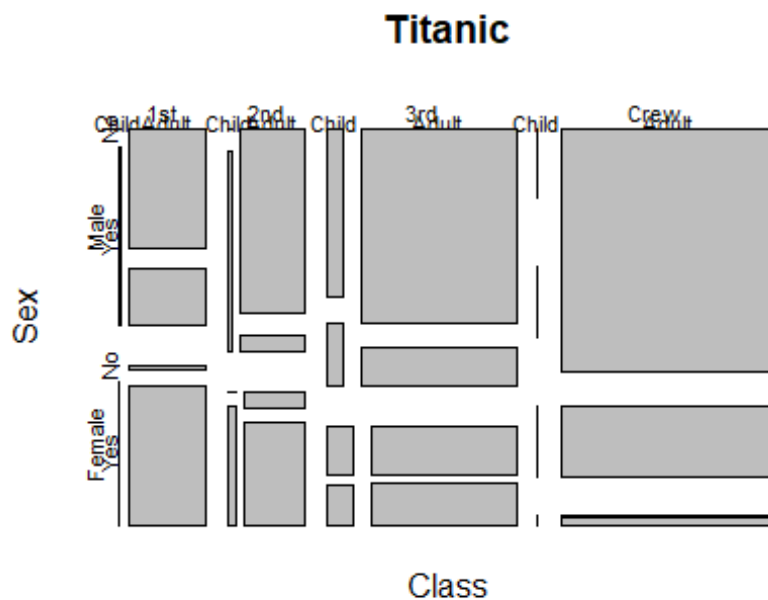
# 繪製 table 資料

# Mosaic Plot (馬賽克圖)是一種專門用來呈現多維類別資料 (Multivariate Categorical Data)的圖形。

# 馬賽克圖用來呈現 32 種組合的人數。矩形越大，代表該組合的人數越多。

# Crew > 3rd (三等艙) > 1st(頭等艙) > 2nd(二等艙)

```
plot(Titanic)
```



# 兒童存活率較高嗎？

```
apply(Titanic, c(3, 4), sum)
```

```
##      Survived
## Age      No Yes
```

```

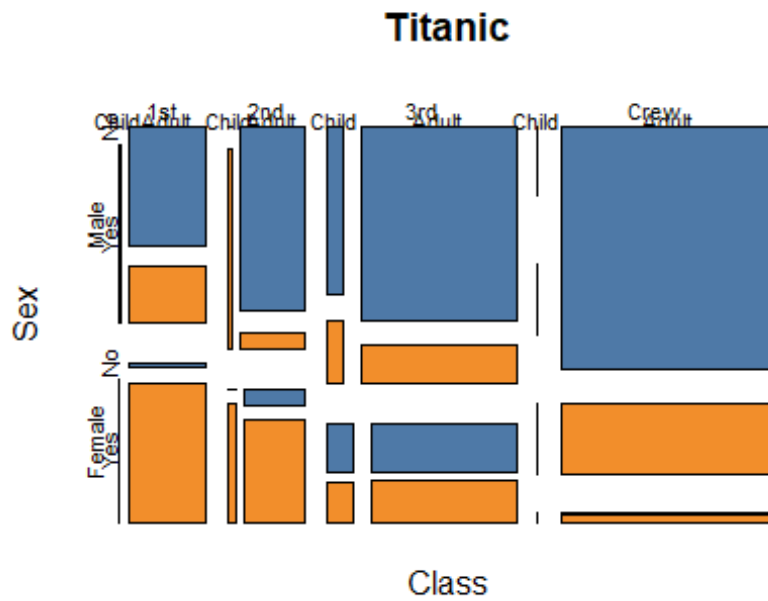
##      Child    52  57
##      Adult 1438 654

# 女性的存活率更高?
apply(Titanic, c(2, 4), sum)

##           Survived
## Sex           No Yes
##   Male    1364 367
##   Female   126 344

# 使用 mosaicplot() 自訂顏色
mosaicplot(
  Titanic,
  color = c(
    "#4E79A7", # 藍
    "#F28E2B", # 橘
    "#59A14F", # 綠
    "#E15759", # 紅
    "#76B7B2", # 青
    "#EDC948", # 黃
    "#B07AA1", # 紫
    "#FF9DA7"  # 粉
  ),
  main = "Titanic",
  xlab = "Class",
  ylab = "Sex"
)

```



# 依 Survival 著色

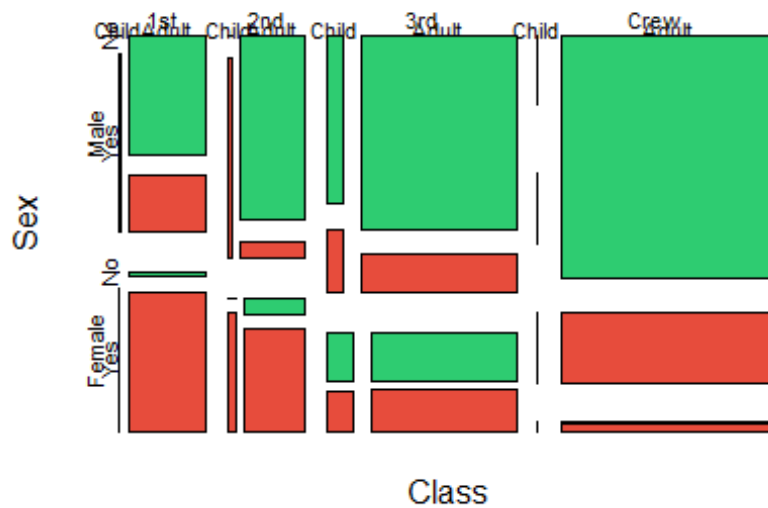
# 生還 (Yes): 綠色

# 死亡 (No) : 紅色

```
cols <- c(
  "#2ECC71", # 生還
  "#E74C3C"  # 死亡
)
```

```
mosaicplot(
  Titanic,
  color = cols,
  main = "Titanic Mosaic Plot with Survival",
  xlab = "Class",
  ylab = "Sex"
)
```

## Titanic Mosaic Plot with Survival

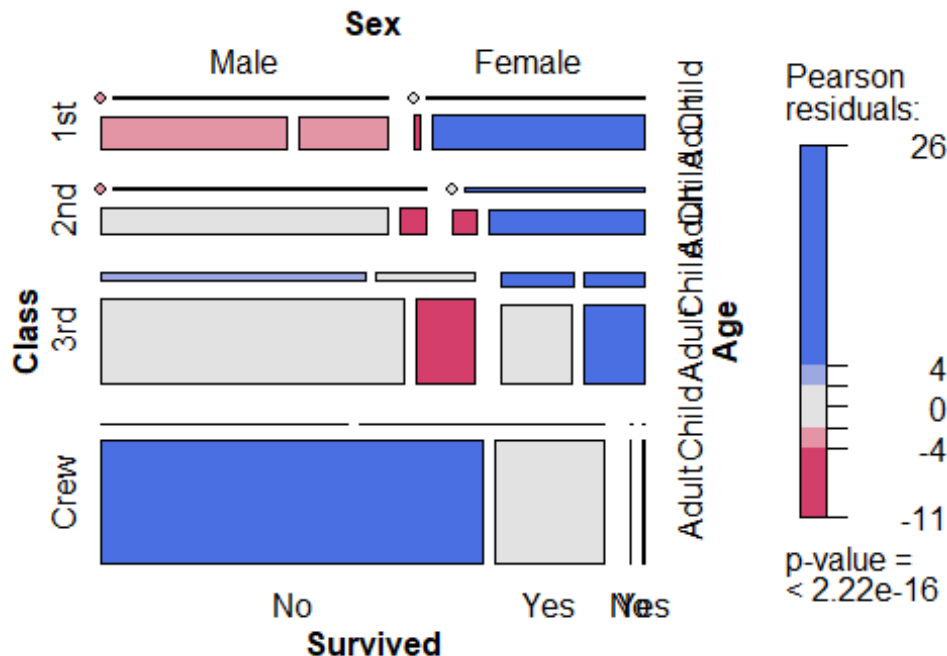


```
# 使用 vcd 套件
# install.packages("vcd")
library(vcd)

## Loading required package: grid

mosaic(
  Titanic,
  shade = TRUE,
  legend = TRUE,
  main = "Titanic Mosaic Plot with vcd package"
)
```

## Titanic Mosaic Plot with vcd package

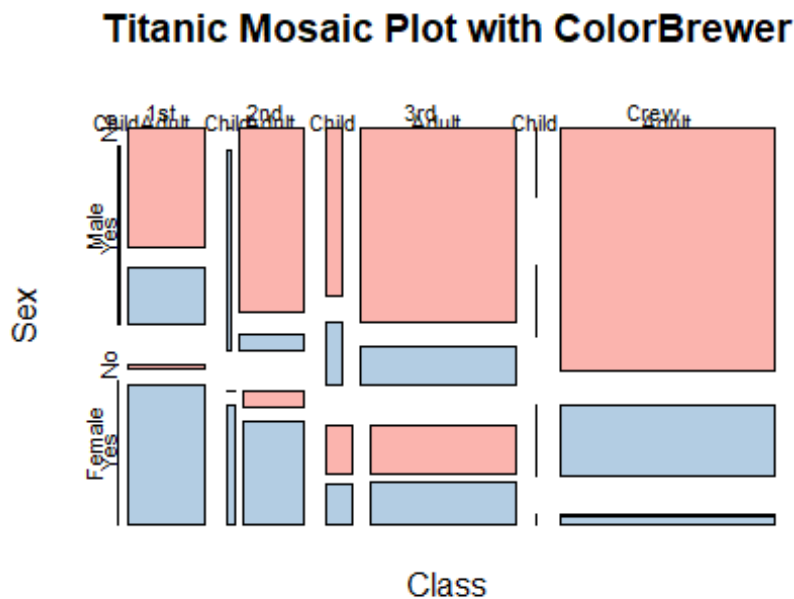


- # 左側 (Y 軸) Class, 依序為: 1st, 2nd, 3rd, Crew, 每一列代表一種艙等。
- # 上方 Sex, 區分成: Male, Female, 左右兩區比較男女差異。
- # 下方 Survived, 區分成: No, Yes, 可以直接比較 死亡 vs 生還。
- # 最右側 Age, 區分成: Adult, Child。
- # 右側圖例顏色代表 Pearson residuals, 即 Pearson 標準化殘差, 表示'實際人數'與'理論期望人數'的差異。
- # 藍色表示 實際人數 > 期望值人數, Observed > Expected. 例如: 頭等艙女性生還, Crew 男性死亡 都是深藍色。
- # 紅色表示 實際人數 < 期望值人數, Observed < Expected. 例如: 頭等艙男性生還是紅色。
- # 灰色表示 實際人數 大約等於 期望值人數, Observed  $\approx$  Expected, 即沒有顯著差異。
- # 圖例下方表示  $p\text{-value} < 2.22e-16$ , 卡方獨立性檢定 (Chi-square Test of Independence) 的檢定結果。
- # 其意義為:
  - # 虛無假設 ( $H_0$ ): Class、Sex、Age 與 Survived 彼此獨立, 沒有關聯。
  - # 對立假設 ( $H_1$ ): Class、Sex、Age 與 Survived 彼此有關聯。
- # 檢定結果:  $p\text{-value}$  極小 (遠小於 0.05), 因此拒絕虛無假設。
- # 檢定結論: 艙等(Class), 性別(Sex), 年齡(Age) 與是否生還(Survived)之間存在顯著關聯。
- # 第1列結果:
  - # 1. 男性區塊比女性寬, 表示頭等艙男性乘客較多。
  - # 2. 女性的 Yes(生還) 區塊遠大於 No (死亡)。

```
# 3. 男性的 No (死亡) 區塊仍相當大.
# 4. 幾乎全部都是 Adult (成人), 兒童非常少.
# 5. Female No 深紅: 死亡遠低於期望值.
# 6. Female Yes 深藍: 生還遠高於期望值.
# 7. 第 1 列結論: 男性沒有明顯偏向生還或死亡, 女性呈現極端結果—幾乎都生還, 很少死亡.
```

```
# 使用 ColorBrewer 配色
library(RColorBrewer)
```

```
mosaicplot(
  Titanic,
  # color = brewer.pal(8, "Set2"),
  color = brewer.pal(8, "Pastel1"),
  main = "Titanic Mosaic Plot with ColorBrewer"
)
```



```
# 將 table 轉換為 data.frame
titanic_df <- as.data.frame(Titanic)
titanic_df
```

### 3. airquality 1973 年紐約空氣品質資料集

```
# ?airquality # 153*6
```

```
data(airquality)
```

```

airquality # 153*6

class(airquality)

## [1] "data.frame"

str(airquality)

## 'data.frame': 153 obs. of 6 variables:
## $ Ozone : int 41 36 12 18 NA 28 23 19 8 NA ...
## $ Solar.R: int 190 118 149 313 NA NA 299 99 19 194 ...
## $ Wind : num 7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
## $ Temp : int 67 72 74 62 56 66 65 59 61 69 ...
## $ Month : int 5 5 5 5 5 5 5 5 5 5 ...
## $ Day : int 1 2 3 4 5 6 7 8 9 10 ...

# Ozone : Ozone (ppb) : 平均臭氧濃度
# Solar.R : Solar R (Lang) : 太陽輻射頻率
# Wind : Wind (mph) : 平均風速(英里/小時)
# Temp : Temperature (degrees F) : 每日最高溫(華氏)
# Month : Month (1--12) : 月份
# Day : Day of month (1--31) : 日期

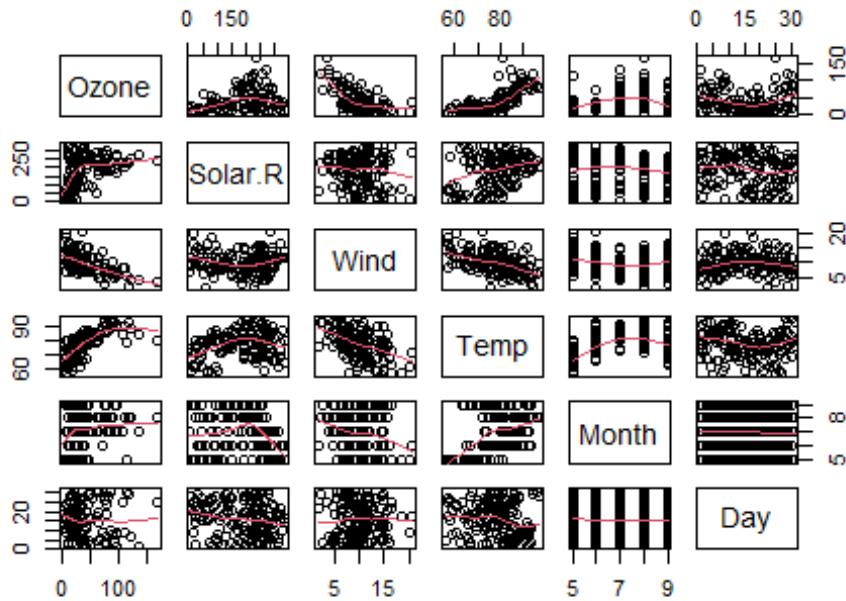
# 有 NA 值
summary(airquality)

## Ozone Solar.R Wind Temp
## Min. : 1.00 Min. : 7.0 Min. : 1.700 Min. :56.00
## 1st Qu.: 18.00 1st Qu.:115.8 1st Qu.: 7.400 1st Qu.:72.00
## Median : 31.50 Median :205.0 Median : 9.700 Median :79.00
## Mean : 42.13 Mean :185.9 Mean : 9.958 Mean :77.88
## 3rd Qu.: 63.25 3rd Qu.:258.8 3rd Qu.:11.500 3rd Qu.:85.00
## Max. :168.00 Max. :334.0 Max. :20.700 Max. :97.00
## NAs :37 NAs :7
## Month Day
## Min. :5.000 Min. : 1.0
## 1st Qu.:6.000 1st Qu.: 8.0
## Median :7.000 Median :16.0
## Mean :6.993 Mean :15.8
## 3rd Qu.:8.000 3rd Qu.:23.0
## Max. :9.000 Max. :31.0
##

# 散佈圖矩陣 (Scatterplot Matrices)
pairs(airquality,
      panel = panel.smooth,
      main = "1973 年紐約空氣品質-散佈圖矩陣")

```

## 1973年紐約空氣品質-散佈圖矩陣



# 找出樣式 (Patterns)

### 4. iris 鳶尾花資料集

# ?iris

`data(iris)`

iris # 150\*5

`str(iris)` # Species 為 factor 物件, 有3種花 ("setosa", "versicolor", "virginica")

```
## 'data.frame': 150 obs. of 5 variables:
## $ Sepal.Length: num 5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
## $ Sepal.Width : num 3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
## $ Petal.Length: num 1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
## $ Petal.Width : num 0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
## $ Species : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1
## 1 1 1 1 1 1 1 ...
```

`levels(iris$Species)`

```
## [1] "setosa" "versicolor" "virginica"
```

`summary(iris)` # 沒有 NA 值, 3種花的樣本數皆各為50筆.

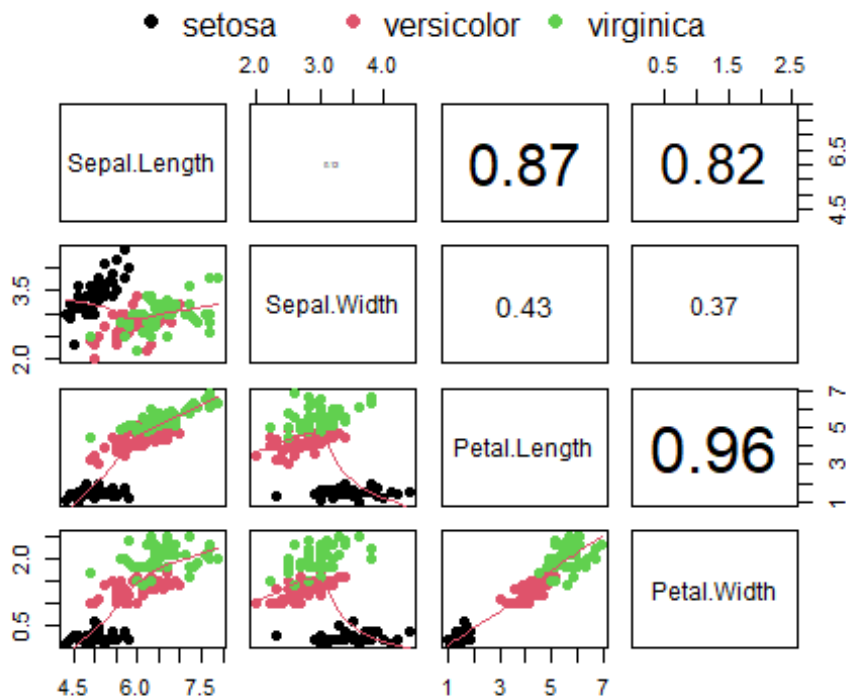


```
## Sepal.Length Sepal.Width Petal.Length Petal.Width
## Min. :4.300 Min. :2.000 Min. :1.000 Min. :0.100
## 1st Qu.:5.100 1st Qu.:2.800 1st Qu.:1.600 1st Qu.:0.300
## Median :5.800 Median :3.000 Median :4.350 Median :1.300
## Mean :5.843 Mean :3.057 Mean :3.758 Mean :1.199
## 3rd Qu.:6.400 3rd Qu.:3.300 3rd Qu.:5.100 3rd Qu.:1.800
## Max. :7.900 Max. :4.400 Max. :6.900 Max. :2.500
## Species
## setosa :50
## versicolor:50
## virginica :50
##
##
##

# 相關係數
panel.cor <- function(x, y, digits = 2, prefix = "", cex.cor, ...) {
  par(usr = c(0, 1, 0, 1))
  r <- abs(cor(x, y))
  txt <- format(c(r, 0.123456789), digits = digits)[1]
  txt <- paste0(prefix, txt)
  if(missing(cex.cor)) cex.cor <- 0.8/strwidth(txt)
  text(0.5, 0.5, txt, cex = cex.cor * r)
}

# iris-散佈圖矩陣
pairs(iris[1:4],
      col = iris$Species,
      pch = 16,
      lower.panel = panel.smooth, # 使用 Lowess
      upper.panel = panel.cor)

legend(x = 0.5,
       y = 1,
       levels(iris$Species),
       pch = 16,
       col = (1:3),
       bty = "n",
       horiz = TRUE,
       xjust = 0.5,
       yjust = 0.2,
       xpd = TRUE)
```



```
# bty = "n" 取消圖例外框
# xpd = TRUE: 所有繪圖都會被裁切至圖形區域(figure region)
# xpd = FALSE: 所有繪圖都會被裁切至繪圖(plot region)
# 找出樣式 (Patterns)
```

## 5. mtcars 汽車資料集

```
# ?mtcars

# mpg      : Miles/(US) gallon           ; 英里/加侖
# cyl      : Number of cylinders         ; 汽缸數
# disp     : Displacement (cu.in.)       ; 排氣量
# hp       : Gross horsepower            ; 總馬力
# drat     : Rear axle ratio              ; 後軸傳動比
# wt       : Weight (1000 lbs)            ; 重量
# qsec     : 1/4 mile time                ; 加速至 1/4 英哩時間
# vs       : Engine (0 = V-shaped, 1 = straight) ; 引擎排列
# am       : Transmission (0 = automatic, 1 = manual); 變速箱(0=自排, 1=手排)
# gear     : Number of forward gears      ; 前進檔數
# carb     : Number of carburetors        ; 化油器數量
data(mtcars)

mtcars # 32*11

str(mtcars) # 所有資料皆為數值型
```

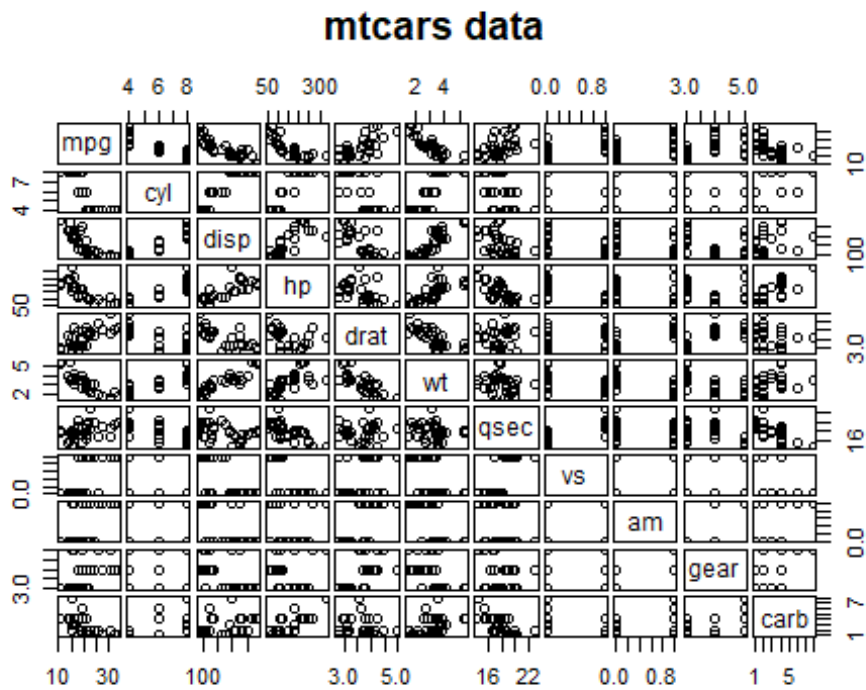
```
## 'data.frame':    32 obs. of  11 variables:
## $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
## $ cyl : num   6  6  4  6  8  6  8  4  4  6 ...
## $ disp: num  160 160 108 258 360 ...
## $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
## $ drat: num   3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
## $ wt  : num   2.62 2.88 2.32 3.21 3.44 ...
## $ qsec: num   16.5 17 18.6 19.4 17 ...
## $ vs  : num    0  0  1  1  0  1  0  1  1  1 ...
## $ am  : num    1  1  1  0  0  0  0  0  0  0 ...
## $ gear: num    4  4  4  3  3  3  3  4  4  4 ...
## $ carb: num    4  4  1  1  2  1  4  2  2  4 ...
```

```
summary(mtcars) # 資料沒有 NA
```

```
##      mpg          cyl          disp          hp
## Min.   :10.40    Min.   :4.000    Min.   : 71.1    Min.   : 52.0
## 1st Qu.:15.43    1st Qu.:4.000    1st Qu.:120.8    1st Qu.: 96.5
## Median :19.20    Median :6.000    Median :196.3    Median :123.0
## Mean   :20.09    Mean   :6.188    Mean   :230.7    Mean   :146.7
## 3rd Qu.:22.80    3rd Qu.:8.000    3rd Qu.:326.0    3rd Qu.:180.0
## Max.   :33.90    Max.   :8.000    Max.   :472.0    Max.   :335.0
##      drat          wt          qsec          vs
## Min.   :2.760    Min.   :1.513    Min.   :14.50    Min.   :0.0000
## 1st Qu.:3.080    1st Qu.:2.581    1st Qu.:16.89    1st Qu.:0.0000
## Median :3.695    Median :3.325    Median :17.71    Median :0.0000
## Mean   :3.597    Mean   :3.217    Mean   :17.85    Mean   :0.4375
## 3rd Qu.:3.920    3rd Qu.:3.610    3rd Qu.:18.90    3rd Qu.:1.0000
## Max.   :4.930    Max.   :5.424    Max.   :22.90    Max.   :1.0000
##      am          gear          carb
## Min.   :0.0000    Min.   :3.000    Min.   :1.000
## 1st Qu.:0.0000    1st Qu.:3.000    1st Qu.:2.000
## Median :0.0000    Median :4.000    Median :2.000
## Mean   :0.4062    Mean   :3.688    Mean   :2.812
## 3rd Qu.:1.0000    3rd Qu.:4.000    3rd Qu.:4.000
## Max.   :1.0000    Max.   :5.000    Max.   :8.000
```

```
# mtcars-散佈圖矩陣
```

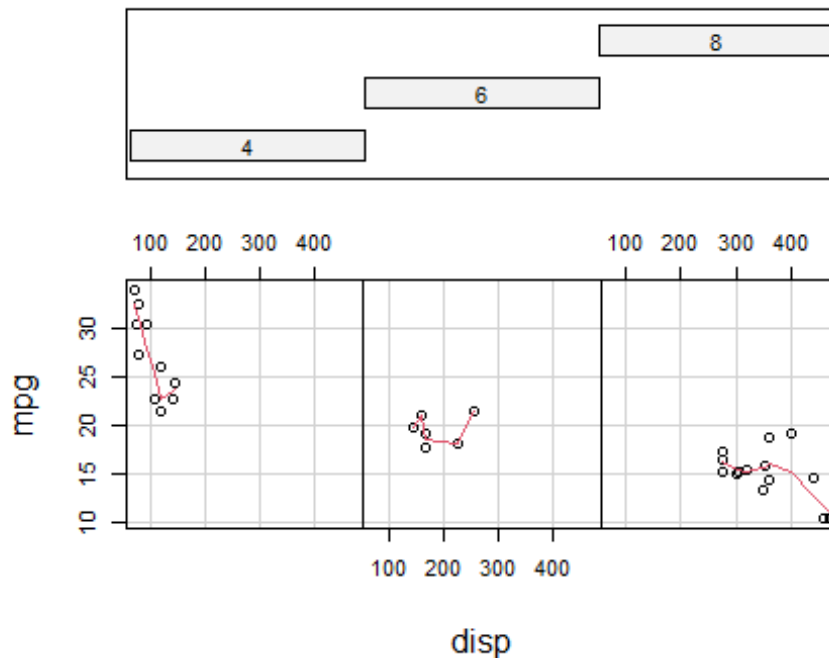
```
pairs(mtcars,
      main = "mtcars data",
      gap = 1/4)
```



# *gap*: 對角線上的變數名稱(*Label*)與左右散佈圖保留多少空白距離, 本例將預設留白縮小為 0.25, 整張圖會更緊湊.

```
# mtcars-條件圖(區分不同 cyl 條件)
coplot(mpg ~ disp | as.factor(cyl),
       data = mtcars,
       panel = panel.smooth,
       rows = 1)
```

Given : as.factor(cyl)



## 6. penguins 南極洲帕爾默站企鵝

```
# s?penguins
```

```
# species: factor, with levels Adelie, Chinstrap, and Gentoo; 3 個物種:
# 阿德利企鵝(Adelie penguin), 頰帶企鵝(chinstrap penguin), 巴布亞企鵝(Gento
# o penguin)
# island: factor, with levels Biscoe, Dream, and Torgersen) ; 島嶼
# bill_len: numeric, bill length (millimeters) ; 喙長(公釐,
# mm)
# bill_dep: numeric, bill depth (millimeters) ; 喙深
# flipper_len: integer, flipper length (millimeters) ; 鰭長
# body_mass: integer, body mass (grams) ; 重量
# sex: factor, with levels female and male ; 性別
# year: integer, study year: 2007, 2008, or 2009 ; 研究年份
```

```
data(penguins)
```

```
penguins # 344*8
```

```
str(penguins) # 資料集包括 3 種不同資料型態: Factor, num, int
```

```
## 'data.frame': 344 obs. of 8 variables:
## $ species : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ island : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3
```

```

3 3 3 3 ...
## $ bill_len : num 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42
...
## $ bill_dep : num 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2
...
## $ flipper_len: int 181 186 195 NA 193 190 181 195 193 190 ...
## $ body_mass : int 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250
...
## $ sex : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2
NA NA ...
## $ year : int 2007 2007 2007 2007 2007 2007 2007 2007 2007 20
07 ...

```

**summary**(penguins) # 有 NA 值

```

##      species      island      bill_len      bill_dep
## Adelie   :152 Biscoe   :168 Min.    :32.10 Min.    :13.10
## Chinstrap: 68 Dream    :124 1st Qu.:39.23 1st Qu.:15.60
## Gentoo   :124 Torgersen: 52 Median  :44.45 Median :17.30
##                                     Mean    :43.92 Mean    :17.15
##                                     3rd Qu.:48.50 3rd Qu.:18.70
##                                     Max.    :59.60 Max.    :21.50
##                                     NAs     :2     NAs     :2
## flipper_len  body_mass      sex      year
## Min.    :172.0 Min.    :2700 female:165 Min.    :2007
## 1st Qu.:190.0 1st Qu.:3550 male   :168 1st Qu.:2007
## Median :197.0 Median :4050 NAs    : 11 Median :2008
## Mean    :200.9 Mean    :4202          Mean    :2008
## 3rd Qu.:213.0 3rd Qu.:4750          3rd Qu.:2009
## Max.    :231.0 Max.    :6300          Max.    :2009
## NAs     :2     NAs     :2

```

**names**(penguins)

```

## [1] "species"      "island"      "bill_len"    "bill_dep"    "flipper
_len"
## [6] "body_mass"    "sex"         "year"

```

# 各欄位之遺漏值數量

**colSums**(is.na(penguins))

```

##      species      island      bill_len      bill_dep flipper_len      body_m
ass
##           0           0           2           2           2
2
##      sex      year
##      11         0

```

# **install.packages**("GGally")

**library**(GGally)

```

ggpairs(
  penguins,
  columns =
    c("bill_len",
      "bill_dep",
      "flipper_len",
      "body_mass"),
  aes(color=species)
)

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).

## Warning: Removed 2 rows containing missing values
## Removed 2 rows containing missing values
## Removed 2 rows containing missing values

## Warning: Removed 2 rows containing missing values or values outside
the scale range
## (`geom_point()`).

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).

## Warning: Removed 2 rows containing missing values
## Removed 2 rows containing missing values

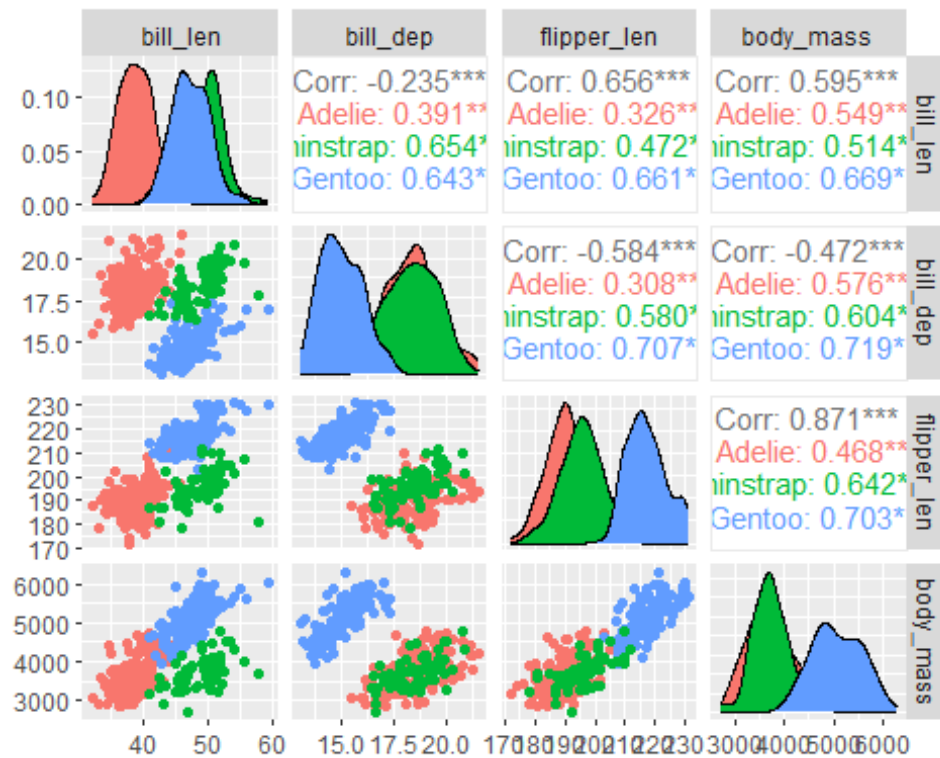
## Warning: Removed 2 rows containing missing values or values outside
the scale range
## (`geom_point()`).
## Removed 2 rows containing missing values or values outside the scale
range
## (`geom_point()`).

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).

## Warning: Removed 2 rows containing missing values
## Warning: Removed 2 rows containing missing values or values outside
the scale range
## (`geom_point()`).
## Removed 2 rows containing missing values or values outside the scale
range
## (`geom_point()`).
## Removed 2 rows containing missing values or values outside the scale
range
## (`geom_point()`).

```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_density()`).
```



```
# 找出樣式 (Patterns)
```

```
end
```